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INTRODUCTION

Orbit prediction builds upon orbit determination by extending the estimated satellite state into the future. Once the position and velocity of a satellite are accurately determined at a given reference time, these values are used as the starting point to forecast its future motion. Accurate orbit prediction is essential for routine mission activities such as ground station contact planning, collision avoidance, manoeuvre scheduling, and long-term mission analysis.

In real-world operations, predicting a satellite's orbit is challenging because its motion is continuously affected by various perturbing forces. Effects such as Earth's non-uniform gravity field, atmospheric drag, and other environmental influences cause the satellite to gradually deviate from an ideal path. As a result, realistic prediction models must account for these forces to ensure that the predicted trajectory remains close to the satellite's actual motion, especially over longer time periods.

In this project, orbit prediction is carried out by propagating the estimated state vectors forward in time using a MATLAB-based computational approach. Starting from the initial state at a known epoch, the orbit is predicted by progressively including perturbations—beginning with ideal two-body motion, followed by the J_2 effect and atmospheric drag. This step-by-step approach clearly demonstrates how each physical effect influences the satellite's future trajectory.

The predicted orbits are then compared with results obtained from STK to assess the accuracy of the developed model. The analysis shows that incorporating realistic perturbation effects significantly improves prediction accuracy. Overall, this work highlights the importance of accurate modelling and numerical propagation in achieving reliable orbit prediction for operational satellite missions.

OBJECTIVES

Primary Objective

To develop and assess reliable methods for predicting a satellite's position and velocity over time by using previously recorded measurements.

Research-Focused Objectives

1. Understanding Orbital Motion

To study how a satellite moves both in an ideal, disturbance-free environment and under real conditions where multiple forces act on its orbit. This helps build a clear distinction between basic orbital motion and the complexities introduced by real-world effects.

2. Studying Perturbation Effects

To explore how different external forces gradually alter a satellite's path over time, focusing on:

Earth's J_2 effect (oblateness)

Atmospheric drag

This allows a deeper understanding of how each factor shapes long-term orbital behaviour.

3. Improving Prediction Methods

To enhance the accuracy and efficiency of the techniques used to propagate state vectors, making sure that the predictions remain dependable while keeping the computations practical for real mission operations.

OVERVIEW OF THEORETICAL FRAMEWORK

A comprehensive literature review was carried out to identify the theoretical principles relevant to the development of the orbital models used in this work. The concepts presented in this section synthesize the key findings from established astrodynamics references and provide the foundational framework upon which the subsequent detailed analysis is built. The topics are organised systematically into three principal layers—**Ideal Orbit, Orbit with J_2 Perturbation, and Orbit with J_2 combined with Atmospheric Drag**—and the corresponding theoretical formulations adopted from the literature are briefly outlined below.

1. Ideal Keplerian Orbit

The first part of the study formulates the ideal orbital model, in which the satellite and the Earth are treated as point masses governed solely by Newtonian gravitational attraction. This model forms the fundamental baseline of orbital mechanics and serves as the reference solution for analysing the effects of perturbations.

Understanding the ideal orbital model requires familiarity with the following foundational concepts:

- **Kepler’s Laws of Planetary Motion**

These laws describe the geometric and temporal characteristics of orbital motion. A clear understanding of these principles is essential for interpreting basic orbital behaviour and for establishing the foundation upon which more advanced models are built.

- **Two-Body Problem**

Provides the analytical solution to central-force motion and yields the ideal Keplerian orbit, forming the mathematical baseline for all perturbation models used in later sections.

- **Constants of Motion**

- *Conservation of Mechanical Energy*: Defines the relationship between orbital speed and size through the vis-viva equation, which is vital for assessing how energy variations influence orbital parameters.

- *Conservation of Angular Momentum*: Determines the geometry and orientation of the orbital plane, forming a key reference for understanding changes caused by perturbing forces.

- **Kepler’s First Law and Trajectory Equations**

These equations mathematically describe the satellite’s path in its orbital plane. Their importance lies in enabling accurate orbit propagation and providing a reference trajectory against which perturbed motion is compared.

- **Conic Sections**

Orbital trajectories follow conic geometries. Knowledge of conic sections is fundamental for analysing orbital shape, eccentricity, and the geometric impact of perturbations.

- **Classical Orbital Elements ($a, e, i, \Omega, \omega, \nu$)**

Provide a complete parametric description of an orbit’s size, shape, orientation, and position, and serve as the reference parameters used to quantify perturbations.

This idealised orbital model establishes the mathematical groundwork required for all subsequent analyses. Both J_2 and atmospheric drag effects are introduced as deviations from this baseline solution.

The description of the Ideal Two-Body Orbit in this section follows the theoretical development given in [1] *Bate, Mueller & White (Chapter 1)*, [2] *Vallado (Chapter 1)*, and [3] *Chobotov (Chapter 3)*, with the relevant material drawn from the chapters in which these authors define the Keplerian framework and establish the governing equations of unperturbed orbital motion.

2. Orbit with J_2 Perturbation

The second component introduces the J_2 perturbation, which accounts for the Earth's oblateness. Due to the equatorial bulge, the Earth's gravity field deviates from spherical symmetry, generating long-term (secular) variations in key orbital elements such as RAAN, argument of perigee, and mean anomaly. This section derives and analyses these secular drift rates.

To understand J_2 -perturbed motion, the following background knowledge is required:

- **Ideal Orbit Theory**

Since J_2 effects are formulated as corrections to the two-body model, a strong grasp of the ideal orbit is essential for discerning how the gravitational asymmetry modifies the natural Keplerian motion.

- **Earth's Gravity Field Representation**

Understanding spherical harmonics particularly the J_2 term is crucial for modelling Earth's gravitational irregularities. This knowledge allows accurate prediction of orientation changes in the orbital plane and perigee.

- **Role of Orbital Elements**

Because J_2 primarily affects Ω , ω , and M , familiarity with the physical meaning of each orbital element is necessary for interpreting changes in plane orientation and perigee rotation.

Introducing J_2 allows the model to reflect the dominant gravitational perturbation in Low Earth Orbit. This enables the study of how the orbital plane and perigee evolve due to the Earth's oblateness.

The analysis of the J_2 -induced gravitational perturbation is based on the classical development found in [2] *Vallado (Chapter 8)*, [3] *Chobotov (Chapter 8-10)* and [4] *Montenbruck (Chapter 3)* with the relevant formulations taken from the chapters addressing Earth's non-spherical gravity field and the resulting secular changes in orbital elements.

3. Orbit with J_2 + Atmospheric Drag

The final component integrates atmospheric drag along with J_2 perturbations, producing a comprehensive model suitable for analysing Low Earth Orbit (LEO) behaviour. Drag acts as the primary non-conservative force in LEO and results in gradual loss of orbital energy, leading to reductions in semi-major axis, eccentricity, and orbital lifetime.

This component builds upon all earlier concepts and additionally requires:

- **Ideal Orbit and J_2 Perturbation Theory**

A combined understanding of Keplerian motion and J_2 -induced effects is essential for accurately evaluating drag-driven changes, as drag modifies the orbit in conjunction with gravitational perturbations.

- **Atmospheric Density Models (NRLMSISE-00)**

Atmospheric density varies with altitude, time, and solar activity. Knowledge of empirical density models is vital for realistic estimation of drag forces and orbital decay rates.

- **Aerodynamic Drag Force Model**

A clear understanding of drag coefficient, cross-sectional area, mass, ballistic coefficient, and relative velocity is required to compute drag acceleration and assess its impact on orbital parameters.

- **Relative Motion Between Satellite and Rotating Atmosphere**

Drag depends on velocity relative to the rotating Earth's atmosphere. Accounting for this relative motion is essential for accurate force modelling and prediction of decay trends.

- **Energy Dissipation and Orbital Decay Mechanics**

Drag continuously removes orbital energy, causing gradual contraction of the orbit. Understanding this process is key to predicting satellite lifetime, decay characteristics, and operational constraints.

Combining atmospheric drag with the J_2 perturbation provides a more realistic representation of LEO dynamics. This unified model forms the basis for predicting long-term orbital decay and assessing satellite operational lifetime.

The treatment of atmospheric drag effects, integrated with the J_2 perturbation, follows the theoretical and empirical frameworks provided in [3] *Chobotov (Chapter 8) and the NRLMSISE-00 model documentation* which is helpful for determination of density at altitude, where the governing equations for drag acceleration and long-term orbital decay are rigorously established.

ANALYTICAL FOUNDATIONS

Astrodynamics is the study of the motion of man-made objects in space under the influence of natural and artificially induced forces. It deals with predicting and analysing the trajectories of satellites and spacecraft as they interact with gravitational fields, atmospheric drag, and other perturbing effects. This field plays a crucial role in mission design, navigation, and the long-term management of space systems.

Understanding how satellites and celestial bodies move under gravitational influence requires a structured theoretical foundation. The earliest insights into orbital motion came from the work of Johannes Kepler, who formulated empirical laws describing how planets orbit the Sun. These laws not only provided the first accurate model of planetary motion but also laid the groundwork for the mathematical and physical formulations that followed. Before transitioning into advanced orbital dynamics, it is essential to begin with Kepler's laws, as they offer the simplest and most intuitive description of orbital behaviour.

1. Kepler's laws of Planetary motion

1. **Law of Orbits:** The orbit of a planet (or any satellite around a central body) is an ellipse with the central body at one of the two foci.

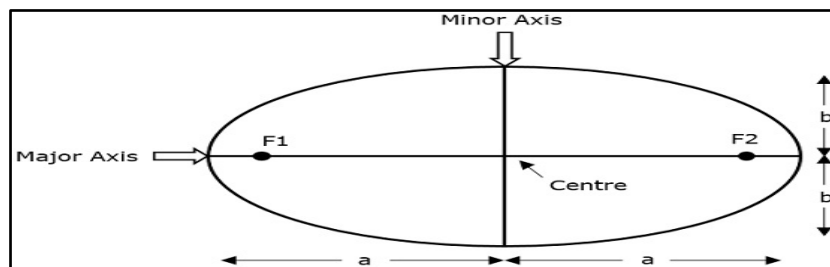


Figure 2: Elliptical Orbit

2. **Law of Areas:** A line segment joining a satellite and the Central body sweeps out equal areas during equal intervals of time.



Figure 3: Sweeping equal area

3. **Law of Periods:** The square of the time period of revolution of a satellite around the central body in an elliptical orbit is directly proportional to the cube of its semi-major axis.

$$\text{Orbital Period } T^2 = \frac{4\pi^2}{GM} a^3$$

The above equation derives the relation between orbital period and semi major axis.

Kepler's laws provide an accurate empirical description of planetary and satellite motion, but they do not explain the physical reasons behind the observed elliptical orbits, the equal-area property, or the precise relationship between orbital size and period. To understand *why* these laws hold true, we must move from observational rules to a fundamental dynamical model based on Newton's law of gravitation. This leads naturally to the two-body problem, where the satellite and central body are treated as point masses interacting only through gravity. By applying Newton's second law to this system, we derive the two-body equation of motion, which mathematically produces the same orbital behaviour summarized by Kepler. In fact, the elliptical trajectory predicted by Kepler's First Law, the area-sweeping relation of the Second Law, and the period–semi-major-axis relation of the Third Law all emerge directly as consequences of the central, inverse-square gravitational force. Thus, the two-body problem provides the physical and mathematical foundation that justifies Kepler's laws and connects classical observations with modern orbital mechanics.

2. Two Body Problem

To develop the two-body equation, we assume:

1. The mass of the satellite is negligible compared to that of the attracting body.
2. The coordinate system used for this problem is inertial, meaning it does not accelerate or rotate. Newton described such an inertial reference frame as being fixed in “absolute space” — something that stays the same and unmoving, no matter what happens around it. However, he never explained how to actually find this “absolutely stationary” frame in reality.

This idea is important because, in calculations, assuming an inertial frame makes things simpler. It means the coordinate system itself isn't changing, so when we take derivatives of vectors (like velocity or acceleration), we don't have to worry about the motion of the coordinate system itself.

3. The bodies of the satellite and attracting body are spherically symmetrical, with uniform density. This allows us to treat each as a point mass.
4. No other forces act on the system except for gravitational forces that act along a line joining the centres of the two bodies.

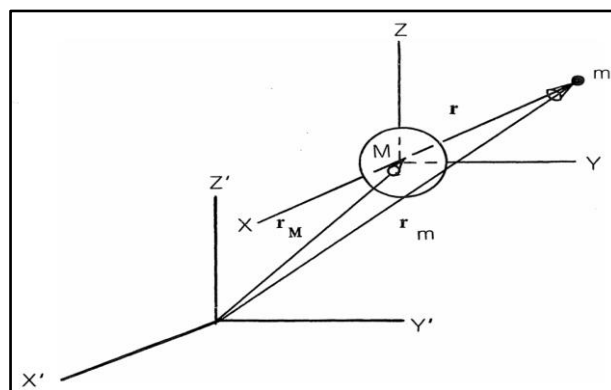


Figure 4: Relative motion of two bodies

Consider the system of two bodies of mass M and m illustrated in the above figure. Let (X', Y', Z') be an inertial set of rectangular cartesian coordinates. Let (X, Y, Z) be a set of nonrotating coordinates parallel to (X', Y', Z') and having an origin coincident with the body of mass M .

The position vectors of the bodies M and m with respect to the set (X', Y', Z') are $\mathbf{r}_M$ and $\mathbf{r}_m$ respectively.

Therefore,

$$\vec{\mathbf{r}} = \mathbf{r}_m - \mathbf{r}_M$$

Now we can apply Newton's law of Gravitation in the inertial frame (X', Y', Z') and obtain

$$m\ddot{\mathbf{r}}_m = -\frac{GMm}{r^2} \frac{\vec{\mathbf{r}}}{r}$$

$$\text{and } M\ddot{\mathbf{r}}_M = \frac{GMm}{r^2} \frac{\vec{\mathbf{r}}}{r}$$

The above equations may be written:

$$\ddot{\mathbf{r}}_m = -\frac{GM}{r^3} \vec{\mathbf{r}}$$

$$\ddot{\mathbf{r}}_M = \frac{Gm}{r^3} \vec{\mathbf{r}}$$

Subtract above equations

$$\ddot{\mathbf{r}} = -\frac{G(M+m)}{r^3} \vec{\mathbf{r}}$$

Note that since the coordinate set (X, Y, Z) is nonrotating with respect to the coordinate set (X', Y', Z'), the magnitudes and directions of r and $\dot{r}$ as measured in the set (X, Y, Z) will be equal respectively to their magnitudes and directions as measured in the inertial set (X', Y', Z').

We will be devoted to studying the motion of artificial satellites, or space probes orbiting about some planet or the sun, the mass of the orbiting body, m , will be much less than that of the central body, M . Hence, we see that

$$G(M+m) \approx GM$$

It is convenient to define a parameter, μ , called the gravitational parameter as

$$\mu \equiv GM$$

The equation becomes

$$\boxed{\ddot{\mathbf{r}} + \frac{\mu}{r^3} \mathbf{r} = \mathbf{0}}$$

The above equation is the two-body equation of motion.

“The relative acceleration between two mutually attracting bodies is directed toward the centre of mass and its magnitude decreases with the square of the distance leading to orbital motion governed by central gravitational force.”

3. Constants of Motion

a) Conservation of Mechanical Energy

When we derive the two-body equation of motion, we see that gravity is the only force acting between the two bodies. Because gravity is a conservative force, it does not add or remove energy from the system—it only changes energy from one form to another. This means that as a satellite moves closer or farther from the central body, its kinetic energy and potential energy may change, but their total always stays the same. This total is called the specific mechanical energy, and it remains constant for any two-body orbit. Therefore, the conservation of mechanical energy follows naturally from the two-body problem and helps us understand how a satellite speeds up near periapsis and slows down near apoapsis.

Thus, using the two-body equation of motion, the conservation of Mechanical Energy is derived as:

1. Dot multiply two body equation of motion by $\dot{\mathbf{r}}$

$$\dot{\mathbf{r}} \cdot \ddot{\mathbf{r}} + \dot{\mathbf{r}} \cdot \frac{\mu}{r^3} \mathbf{r} = 0$$

2. Since in general $\mathbf{a} \cdot \dot{\mathbf{a}} = a\dot{a}$, $\mathbf{v} = \dot{\mathbf{r}}$ and $\dot{\mathbf{v}} = \ddot{\mathbf{r}}$, then

$$v\dot{v} + \frac{\mu}{r^3} r\dot{r} = 0$$

3. Noticing that $\frac{d}{dt}\left(\frac{v^2}{2}\right) = v\dot{v}$ and $\frac{d}{dt}\left(-\frac{\mu}{r}\right) = \frac{\mu}{r^2} \dot{r}$

$$\frac{d}{dt}\left(\frac{v^2}{2}\right) + \frac{d}{dt}\left(-\frac{\mu}{r}\right) = 0$$

$$\text{Or } \frac{d}{dt}\left(\frac{v^2}{2} - \frac{\mu}{r}\right) = 0$$

4. If the time rate of change of an expression is zero, that expression must be a constant which we will call ξ (specific mechanical energy).

$$\xi = \left(\frac{v^2}{2} - \frac{\mu}{r}\right) + c = 0$$

The first term of ξ is obviously the kinetic energy per unit mass of the satellite, the second term is the potential energy per unit mass, which is obtained by equating it with the work done in moving a satellite from one point in space to another against the force of gravity.

The choice of a value for c is arbitrary, allowing us to designate the condition for

$\xi = 0$. The physics community should recognize this equation where $c = \mu/r_{\oplus}$. ξ varies with height and speed and is zero at the Earth's surface.

For simplification,

$$\xi = \frac{v^2}{2} - \frac{\mu}{r} = 0$$

We conclude, therefore, that the specific mechanical energy, ξ , of a satellite which is the sum of its kinetic energy per unit mass and its potential energy per unit mass remains constant along its orbit, neither increasing nor decreasing as a result of its motion.

$$\xi = \frac{v^2}{2} - \frac{\mu}{r} = \frac{h^2}{2r_p^2} - \frac{\mu}{r_p}$$

We already know that $r_p = a(1 - e)$ and $h = \sqrt{\mu p} = \sqrt{\mu a(1 - e^2)}$

$$\xi = \frac{\mu a(1-e^2)}{2a^2(1-e)^2} - \frac{\mu}{a(1-e)}$$

The result (for orbits other than parabolic) is then

$$\xi = -\frac{\mu}{2a}$$

This gives rise to the traditional form of the vis-viva equation.

$$v^2 = \mu \left(\frac{2}{r} - \frac{1}{a} \right)$$

The vis-viva equation is significant because it relates a satellite's velocity to its position and orbit size, allowing the orbital speed to be determined at any point along a Keplerian orbit.

b) Conservation of Angular Momentum

In the two-body problem, gravity always acts along the line joining the satellite and the central body, which means the force has no sideways (tangential) component. Because no torque acts on the satellite, its angular momentum stays constant throughout the motion. This is why the satellite always moves in a fixed plane and why its orbit remains stable unless an external force is applied.

Thus, using the two-body equation of motion, the conservation of Angular Momentum is derived as:

1. Cross multiply two body equation by $\mathbf{r}$

$$\mathbf{r} \times \ddot{\mathbf{r}} + \mathbf{r} \times \frac{\mu}{r^3} \mathbf{r} = 0$$

2. Since in general $\mathbf{r} \times \mathbf{r} = 0$, the second term vanishes and

$$\mathbf{r} \times \ddot{\mathbf{r}} = 0$$

3. Considering the differential equation $\frac{d}{dt}(\mathbf{r} \times \dot{\mathbf{r}}) = \dot{\mathbf{r}} \times \dot{\mathbf{r}} + \mathbf{r} \times \ddot{\mathbf{r}}$

$$\frac{d}{dt}(\mathbf{r} \times \dot{\mathbf{r}}) = 0 \quad \text{or} \quad \frac{d}{dt}(\mathbf{r} \times \mathbf{v}) = 0$$

The expression $\mathbf{r} \times \mathbf{v}$ which must be a constant of the motion is simply the vector $\mathbf{h}$, called specific angular momentum.

$$\mathbf{h} = \mathbf{r} \times \mathbf{v}$$

Since $\mathbf{h}$ is the vector cross product of $\mathbf{r}$ and $\mathbf{v}$ it must always be perpendicular to the plane containing $\mathbf{r}$ and $\mathbf{v}$. But $\mathbf{h}$ is a constant vector so $\mathbf{r}$ and $\mathbf{v}$ must always remain in the same plane. Therefore, *we conclude that the satellite's motion must be confined to a plane which is fixed in space.* We shall refer to this as the orbital plane.

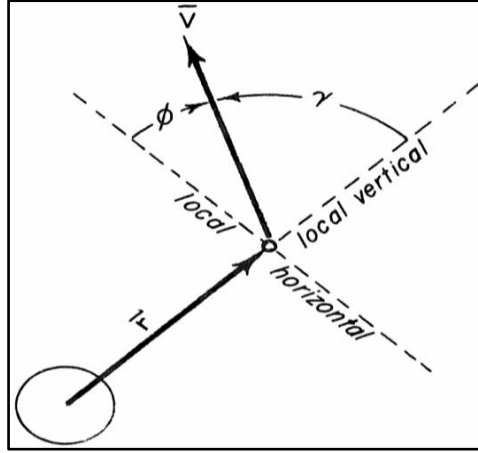


Figure 5: flight path angle

No matter where a satellite is located in space it is always possible to define "up and down" and "horizontal." "Up" simply means away from the centre of the earth and "down" means toward the centre of the earth. So, the local vertical at the location of the satellite coincides with the direction of the vector r . The local horizontal plane must then be perpendicular to the local vertical. We can now define the direction of the velocity vector, v , by specifying the angle it makes with the local vertical as γ , the zenith angle. The angle between the velocity vector and the local horizontal plane is called ϕ , the flight path elevation angle or simply "flight path angle." From the definition of the cross product the magnitude of h is

$$h = rv \sin \gamma \text{ or}$$

$$h = rv \cos \phi$$

The sign of ϕ will be the same as the sign of $r \cdot v$.

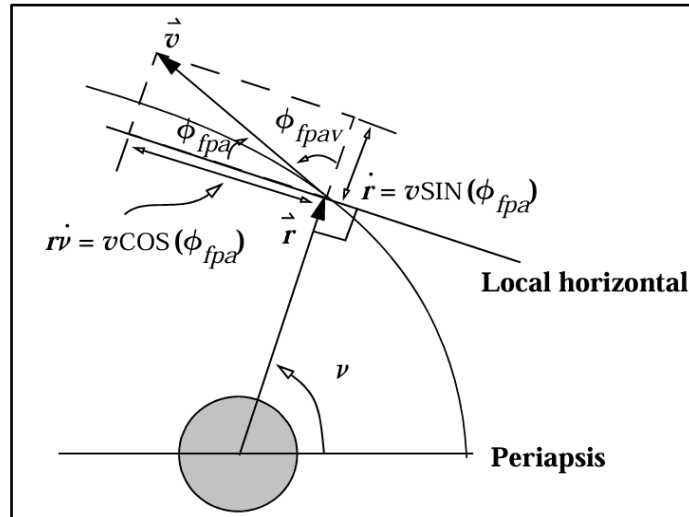


Figure 6: Velocity in terms of change of position and true anomaly

$$v = \dot{r} + r\dot{\nu} \text{ (here } \dot{\nu} \text{ is rate of change of true anomaly)}$$

$$\cos \phi = \frac{r\dot{\nu}}{v}$$

thus,

$$h = r^2 \dot{\nu}$$

This tells us that as a satellite moves closer or farther from Earth, its speed adjusts so that it always sweeps out equal areas in equal time.

4. Kepler's First law (Trajectory Equations)

The trajectory of the satellite—whether it becomes a circle, ellipse, parabola, or hyperbola—is simply the natural result of these two effects working together: the inward gravitational pull shapes how the distance changes, while the conserved angular momentum controls the sideways motion. When these two elements balance, the path that emerges is the familiar conic-section orbit described in orbital mechanics.

The trajectory equation is derived below:

Begin by crossing the two-body equation into the specific angular momentum vector, h :

$$\ddot{\mathbf{r}} \times \mathbf{h} + \frac{\mu}{r^3} \mathbf{r} \times \mathbf{h} = 0$$

Notice the first term is simply the derivative,

$$\frac{d}{dt}(\dot{\mathbf{r}} \times \mathbf{h}) = \ddot{\mathbf{r}} \times \mathbf{h} + \dot{\mathbf{r}} \times \dot{\mathbf{h}} = \ddot{\mathbf{r}} \times \mathbf{h} \text{ because } \dot{\mathbf{h}} = 0$$

$$\frac{\mu}{r^3} (\mathbf{r} \times \mathbf{h}) = \frac{\mu}{r^3} (\mathbf{r} \times (\mathbf{r} \times \mathbf{v})) = \frac{\mu}{r^3} ((\mathbf{r} \cdot \mathbf{v})\mathbf{r} - (\mathbf{r} \cdot \mathbf{r})\mathbf{v}) = \frac{\mu}{r^2} \dot{\mathbf{r}} \mathbf{r} - \frac{\mu}{r} \mathbf{v} = -\mu \frac{d}{dt} \left(\frac{\vec{r}}{r} \right)$$

$$\frac{d}{dt}(\dot{\mathbf{r}} \times \mathbf{h}) = \mu \frac{d}{dt} \left(\frac{\vec{r}}{r} \right)$$

Integrating both the sides,

$$\dot{\mathbf{r}} \times \mathbf{h} = \mu \frac{\vec{r}}{r} + B$$

Now, dot multiply this equation by $\mathbf{r}$ to get a scalar relation:

$$\mathbf{r} \cdot (\dot{\mathbf{r}} \times \mathbf{h}) = \mathbf{r} \cdot \left(\mu \frac{\vec{r}}{r} + B \right)$$

$$\text{Here } (\mathbf{r} \times \dot{\mathbf{r}}) \cdot \mathbf{h} = h^2$$

Considering the dot product angle as true anomaly (v)

$$h^2 = \mu r + r B \cos(v)$$

$$r = \frac{h^2/\mu}{1 + B/\mu \cos(v)}$$

This expression is the polar form of a conic section. B/μ is defined as eccentricity, e , the quantity h^2/μ is equal to the semi-parameter or semi-latus rectum, p .

$$r = \frac{p}{1 + e \cos(v)}$$

It describes the exact shape of a Keplerian orbit, showing how the satellite's distance from Earth changes with its position along the orbit.

5. Conic Sections

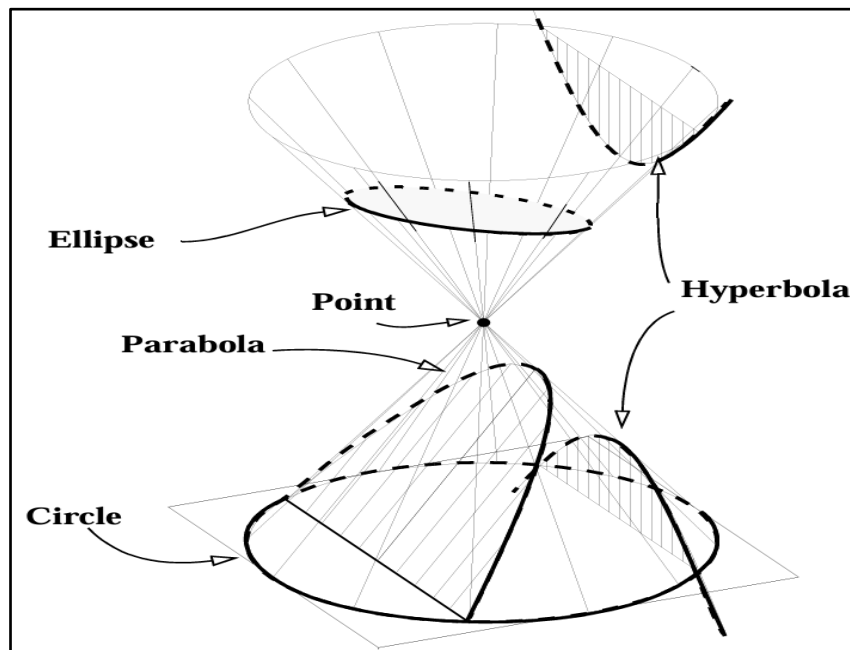


Figure 7: Conic Sections

A cone, when sliced by a plane, produces a family of curves known as **conic sections**.

1. Circle

When the slicing plane is **exactly perpendicular** to the cone's axis of revolution, the intersection forms a perfect circle. This occurs because the plane cuts through the cone symmetrically at a constant distance from the vertex.

2. Ellipse

If the plane cuts the cone at an angle **less steep** than the cone's slope—but not perpendicular—the resulting shape is an ellipse. This is the most general conic section, and circles are simply a special case of ellipses. Most bound orbital trajectories (artificial satellites, planets) follow elliptical paths.

3. Parabola

When the plane is **parallel to the slanted outer edge** (the generator) of the cone, it produces a parabola. This curve represents the boundary between open and closed orbits in orbital mechanics, corresponding to exactly escape velocity.

4. Hyperbola

If the cutting plane is **parallel to the cone's axis** or at an angle steeper than the cone's slope, it intersects both halves of the double cone. This generates a pair of symmetrical open curves called hyperbolas. In celestial mechanics, objects following hyperbolic paths exceed escape velocity and never return.

5. Degenerate Cases

- If the plane passes **through the cone's vertex only**, the intersection collapses to a single point.
- If the plane lies **exactly on the surface of the cone**, the intersection becomes a straight line (**Rectilinear Orbit**).

These conic sections—circle, ellipse, parabola, and hyperbola—form the mathematical foundation of orbital trajectories. Every possible path a satellite or celestial body can follow under gravity corresponds to one of these conic shapes.

To understand and describe the motion of a satellite in space, it is essential to begin with the concept of orbital elements. These elements provide a systematic way to define the size, shape, orientation, and position of an orbit at any given time. Instead of dealing directly with raw position and velocity components, orbital elements offer a more intuitive and physically meaningful representation of the spacecraft's trajectory. Introducing these elements at the beginning establishes a clear foundation for the methodology that follows, as each computational step ultimately relates to how these elements change or are used to reconstruct the satellite's state.

6. Orbital Elements

The set of orbital elements can be divided into two groups: the dimensional elements and the orientation elements.

The dimensional elements specify the size and shape of the orbit and relate the position in the orbit to time

They are as follows:

1. **a = semimajor axis**, which specifies the size of the orbit.
2. **e = eccentricity**, which specifies the shape of the orbit.
3. **ν = true anomaly** or perigee passage, which relates position in orbit to time (ν is often replaced by M , the mean anomaly at some arbitrary time t ; the mean anomaly is a uniformly varying angle)

The orientation elements specify the orientation of the orbit in space.

They are as follows:

4. **i = inclination** of the orbit plane with respect to the reference plane, which is taken to be the Earth's equator plane for satellite orbits. $0 \text{ deg} \leq i \leq 180 \text{ deg}$.
For $0 \text{ deg} \leq i \leq 90 \text{ deg}$, the motion is "posigrade" or "direct"; for $90 \text{ deg} < i \leq 180 \text{ deg}$, the motion is termed "retrograde"
5. **Ω = right ascension** of the ascending node (often shortened to simply "node"); Ω is measured counterclockwise in the equator plane, from the direction of the vernal equinox to the point at which the satellite makes its south-to-north crossing of the equator ($0 \text{ deg} \leq \Omega < 360 \text{ deg}$)
6. **ω = argument of perigee**; ω is measured in the orbit plane in the direction of motion, from the ascending node to perigee ($0 \text{ deg} \leq \omega < 360 \text{ deg}$)

The angles i and Ω specify the orientation of the orbit plane in space. The angle ω then specifies the orientation of the orbit in its plane.

The argument of latitude u defines the position of the satellite relative to the node line. Frequently the **semi-latus rectum**, p , is substituted for a .

Instead of argument of periapsis, the following is sometimes used:

π longitude of periapsis-the angle from I to periapsis measured eastward to the ascending node (if it exists) and then in the orbital plane to periapsis. If both Ω and ω are defined then

$$\pi = \Omega + \omega$$

Any of the following may be substituted for time of periapsis passage and would suffice to locate the satellite at t_0 :

v_0 true anomaly at epoch- the angle, in the plane of the satellite's orbit, between periapsis and the position of the satellite at a particular time, t_0 called the "epoch."

u_0 argument of latitude at epoch-the angle, in the plane of the orbit, between the ascending node and the radius vector to the satellite at time t_0 '.

$$u_0 = \omega + v_0$$

l_0 true longitude at epoch - the angle between I and r_0 (the radius vector to the satellite at t_0 measured eastward to the ascending node and then in the orbital plane to r_0

$$l_0 = \Omega + \omega + v_0 = \pi + v_0 = \Omega + u_0$$

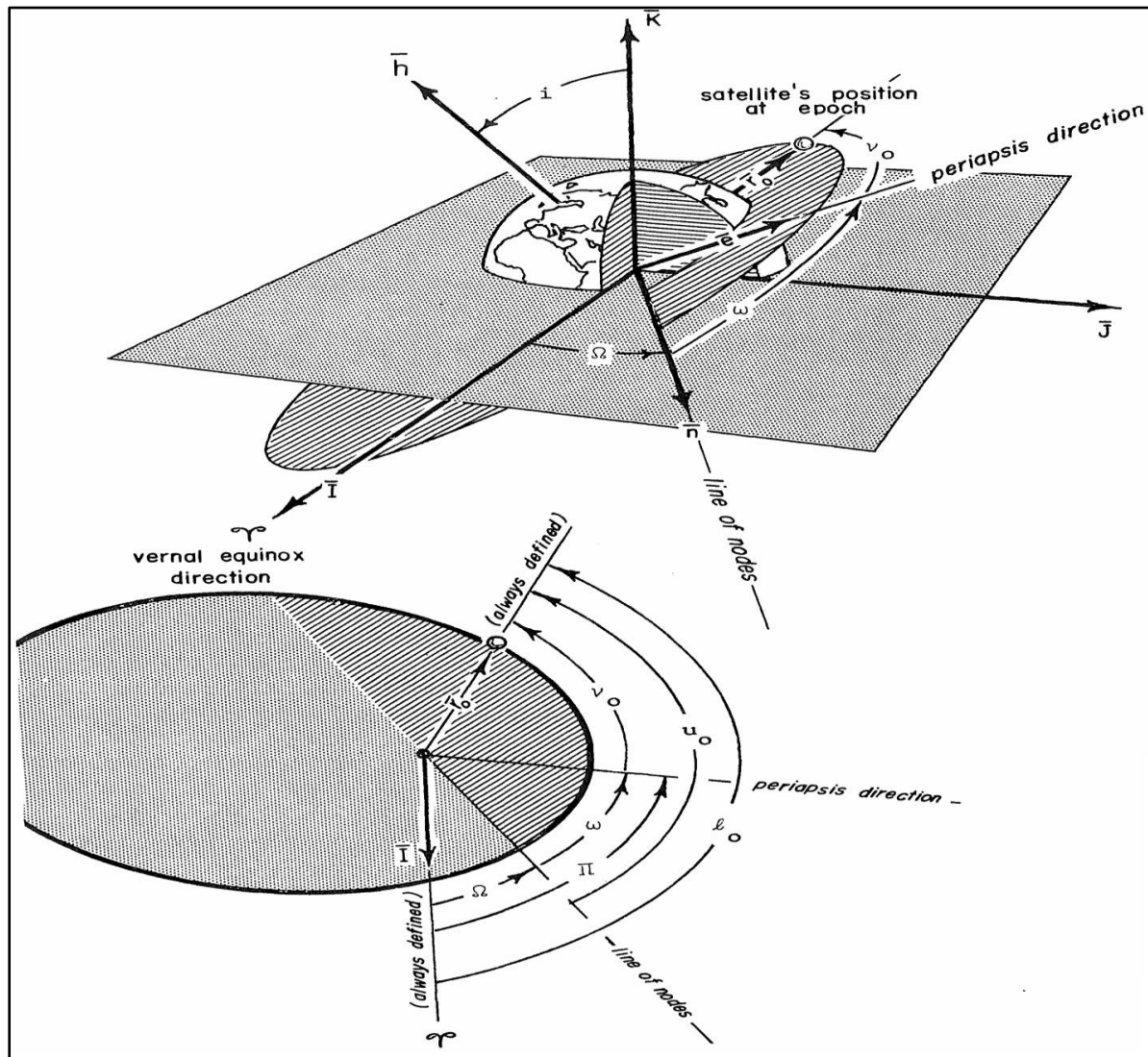


Figure 8: Orbital Elements

METHODOLOGY

Satellite motion can be understood through two main cases: the ideal orbit and the perturbed orbit. In the ideal case, the satellite follows a perfect Keplerian trajectory determined solely by the central body's gravity, resulting in smooth conic-section paths such as circles or ellipses. In contrast, the perturbed orbit accounts for additional influences that slightly alter the satellite's motion over time, causing deviations from the ideal path.

Ideal Keplerian Orbit

a) Determining the orbital elements from $\mathbf{r}$ and $\mathbf{v}$ vectors

Let us assume that a ground station with its antennas can provide us with the vectors $\mathbf{r}$ and $\mathbf{v}$ representing the position and velocity of a satellite relative to the geocentric-equatorial reference frame at a particular time, t_0 .

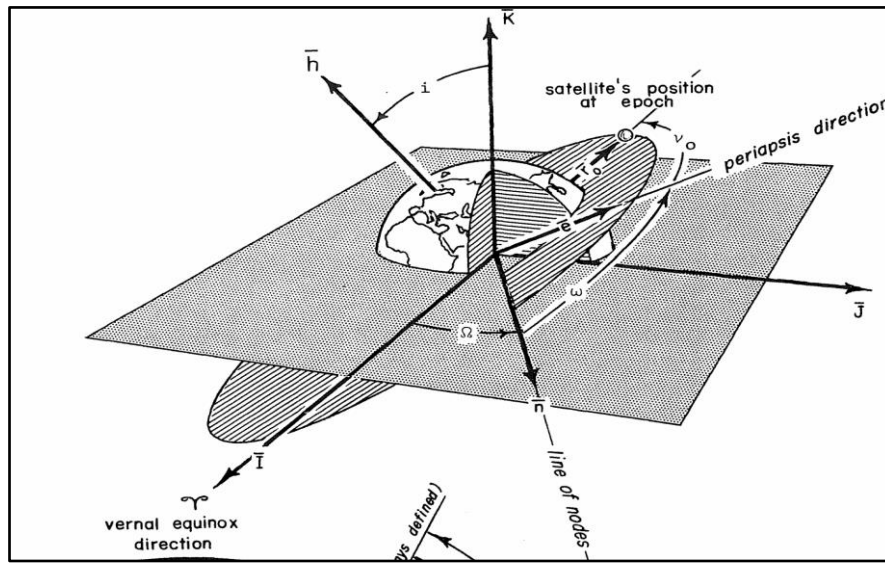


Figure 9: Orbital Elements

The first step is to form the three vectors, $\mathbf{h}$, $\mathbf{n}$ and $\mathbf{e}$.

a) The angular momentum vector $\mathbf{h}$ is given by

$$\mathbf{h} = \mathbf{r} \times \mathbf{v}$$

Thus

$$\mathbf{h} = \begin{vmatrix} \mathbf{I} & \mathbf{J} & \mathbf{K} \\ r_I & r_J & r_K \\ v_I & v_J & v_K \end{vmatrix} = h_I \mathbf{I} + h_J \mathbf{J} + h_K \mathbf{K}$$

$\mathbf{h}$ is a vector perpendicular to the plane of the orbit.

b) The node vector $\mathbf{n}$ is given by

$$\mathbf{n} = \mathbf{K} \times \mathbf{h}$$

$$\text{Thus, } \mathbf{n} = \begin{vmatrix} \mathbf{I} & \mathbf{J} & \mathbf{K} \\ 0 & 0 & 1 \\ h_I & h_J & h_K \end{vmatrix} = n_I \mathbf{I} + n_J \mathbf{J} + n_K \mathbf{K} = -h_J \mathbf{I} + h_I \mathbf{J}$$

$\mathbf{n}$ vector is perpendicular to both $\mathbf{K}$ and $\mathbf{h}$, for which it must lie in both equatorial plane and orbital plane in respective conditions.

Thus, it is considered along their intersection which is called line of nodes and its direction is consider along the ascending node. Only direction is important for our solution.

- c) The vector e is simply the eccentricity of the orbit which points from the focus to periapsis point.

$$\mathbf{e} = \frac{1}{\mu} \left[\left(v^2 - \frac{\mu}{r} \right) \mathbf{r} - (\mathbf{r} \cdot \mathbf{v}) \mathbf{v} \right]$$

- d) Semi latus rectum

$$p = \frac{h^2}{\mu}$$

- e) $e = |\mathbf{e}|$

- f) Since i is the angle between $\mathbf{K}$ and $\mathbf{h}$

$$\cos i = \frac{h_k}{h}$$

inclination i is always less than 180°

- g) Since Ω is the angle between $\mathbf{I}$ and $\mathbf{n}$,

$$\cos \Omega = \frac{n_I}{n}$$

(If $n_I > 0$ then Ω is less than 180°)

- h) Since ω is the angle between $\mathbf{n}$ and $\mathbf{e}$,

$$\cos \omega = \frac{\mathbf{n} \cdot \mathbf{e}}{ne}$$

(If $e_K > 0$ then ω is less than 180°)

- i) Since ν_0 is the angle between $\mathbf{e}$ and $\mathbf{r}$,

$$\cos \nu_0 = \frac{\mathbf{e} \cdot \mathbf{r}}{er}$$

(If $\mathbf{r} \cdot \mathbf{v} > 0$ then ν_0 is less than 180°)

- j) Since u_0 is the angle between $\mathbf{n}$ and $\mathbf{r}$,

$$\cos u_0 = \frac{\mathbf{n} \cdot \mathbf{r}}{nr}$$

(if $r_k > 0$ then u_0 is less than 180°)

- k) $l_0 = \Omega + \omega + \nu_0 = \Omega + u_0$

From above discussions, if $\mathbf{r}$ and $\mathbf{v}$ are known at some time t , we can obtain the orbital elements at that time t . Of which ρ , e , i , Ω , ω are constants for a two-body problem, only true anomaly ν_0 changes with time.

Result 1: Computational results of Orbital elements in MATLAB (Appendix A)

The initial state vectors of the Earth observation satellite (E08) were provided as input to the MATLAB code, which then used the above derived formulas to compute the corresponding orbital elements.

Input State vectors

Input $\mathbf{r}_0 = [4771.810686 \ -3024.529533 \ 3804.492540]$ km

Input $\mathbf{v}_0 = [2.888529 \ 6.849725 \ 1.814147]$ km/s

corresponding computed orbital elements

$a = 6814.901875$ km,

$e = 0.000819$,

$i = 37.367910$ deg

RAAN $\Omega = 265.771134$ deg

omega $\omega = 115.120347$ deg

$\nu_0 = 311.851501$ deg

Period = 5598.873197 s (1.555243 hr)

b) Change in true anomaly

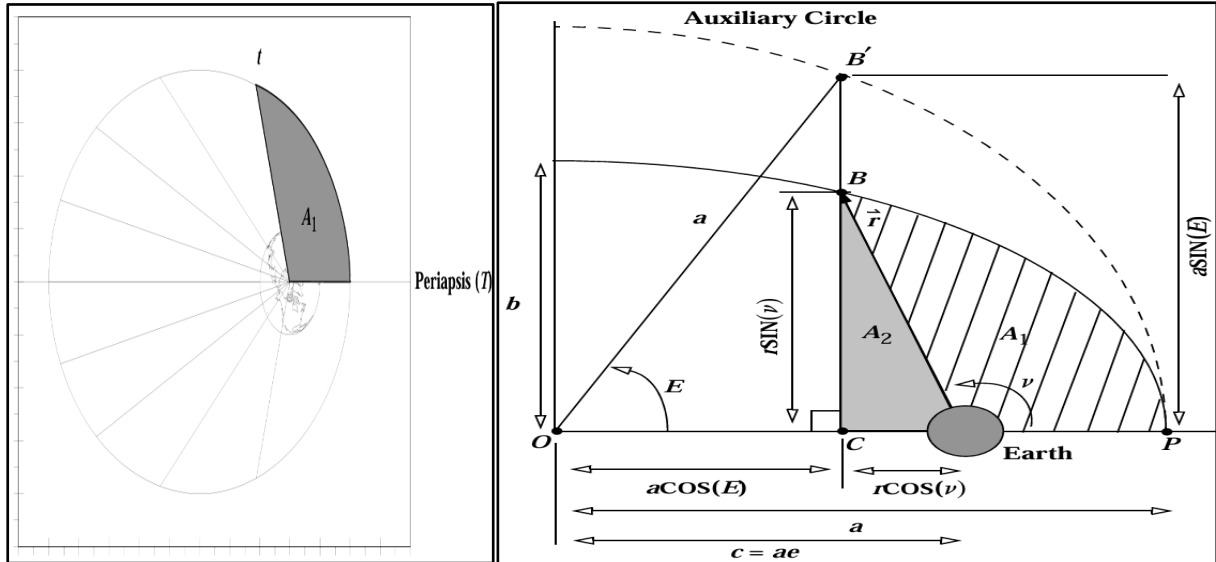


Figure 10: Eccentric Anomaly

Eccentric Anomaly:

The total area of an ellipse is defined by the product of the semi-major and semi-minor axes and pi (πab).

The second law states that equal areas are swept out in equal times,

$$\frac{t-T}{A_1} = \frac{\rho}{\pi ab}$$

where the relation $(t - T)$ represents the change in time, and is the period of one orbital revolution. The only unknown in this equation is the area, A_1 , which we can determine through geometric operations.

The circle drawn around the ellipse is called the auxiliary circle, and the new angle to be defined is the eccentric anomaly, E , which is related to the true anomaly, ν , as shown. It's convenient to express the scaling factor between circle and ellipses.

Let

$$\sqrt{1 - e^2} = \frac{b}{a}$$

To find an expression for the area, A_1 , in terms of the eccentric anomaly,

$$A_1 = \text{Area PCB} - \text{Triangle } A_2$$

Use standard geometric and trigonometric relations to find the area, A_2 :

$$A_2 = \frac{1}{2} (ae - a \cos(E)) \left(\frac{b}{a} \sin(E) \right) = \frac{ab}{2} (\sin(E) - \cos(E) \sin(E))$$

Find the area of the segment of the ellipse, PCB, by using the area of a circle, POB', subtracting the triangle, OB'C, and scaling the result by multiplying by b/a , which relates the vertical distances of an ellipse and a circle. Thus,

$$\begin{aligned} \text{Area PCB} &= \left[\frac{a^2 E}{2} - \frac{1}{2} (a \cos(E)) (a \sin(E)) \right] \frac{b}{a} = \frac{abE}{2} - \frac{a^2}{2} [\cos(E) \sin(E)] \frac{b}{a} \\ &= \frac{ab}{2} [E - \cos(E) \sin(E)] \end{aligned}$$

$$A_1 = \frac{ab}{2} [E - \sin(E)]$$

Substituting in above derived Kepler's second law relation

$$\rho = \frac{2\pi(t-T)}{E - \sin(E)}$$

From Kepler's third law

$$\begin{aligned} \rho &= 2\pi \sqrt{\frac{a^3}{\mu}} \\ \sqrt{\frac{a^3}{\mu}} &= \frac{(t - T)}{E - \sin(E)} \end{aligned}$$

This equation known as **Kepler's equation**, relates the time of flight from periapsis to the eccentric anomaly, semimajor axis, and eccentricity. Kepler then introduces notation for the mean anomaly, M , as

$$M = E - \sin(E) = \sqrt{\frac{\mu}{a^3}} (t - T)$$

and also introduces the mean motion, n , or mean angular velocity. The mean motion is the mean angular rate of the orbital motion as defined in

$$n = \frac{2\pi}{\rho} = \sqrt{\frac{\mu}{a^3}}$$

Kepler's third law relates to the semimajor axis, so n is a function of a .

Mean, eccentric, and true anomaly are all equal at periapsis, $E = M = n = 0^\circ$. Likewise, all three elements are equal to 180° at apoapsis.

$$\mathbf{M} = \mathbf{n}(t - T) = \mathbf{E} - e \sin E \text{ where } \mathbf{n} = \sqrt{\frac{\mu}{a^3}} \text{ and } t - T = \sqrt{\frac{a^3}{\mu}} (\mathbf{E} - e \sin E)$$

Where M is “mean anomaly”, n is called “mean motion” and E is “eccentric anomaly”, whereas $t - T$ is the time of flight.

From the diagram,

$$\sin(E) = \frac{r \sin v}{a \sqrt{1-e^2}} \text{ and } \cos(E) = \frac{ae + r \cos v}{a}$$

To find true anomaly v from E

Substitute $r = \frac{p}{1 + e \cos v} = \frac{a(1-e^2)}{1 + e \cos v}$ in above equations results,

$$\sin(E) = \frac{\sin v \sqrt{1-e^2}}{1 + e \cos v} \text{ and } \cos(E) = \frac{e + \cos v}{1 + e \cos v}$$

For Parabola, $t - T = \frac{1}{2\sqrt{\mu}} \left[pD + \frac{1}{3}D^3 \right]$

Where D is “parabolic eccentric anomaly” and $D = \sqrt{p} \tan \frac{v}{2}$

For Hyperbola, $t - T = \sqrt{\frac{(-a^3)}{\mu}} (e \sinh F - F)$

$$\cosh F = \frac{e + \cos v}{1 + e \cos v}$$

c) Using the obtained values, determination of r and v at any new time

To determine r and v at any new time, we use the obtained $\rho, e, i, \Omega, \omega$ from the above steps and consider the derived true anomaly v to obtain the position and velocity at that time in that orbit. Initially we express position and velocity vector, r and v, obtained from orbital elements, in terms of Perifocal systems (satellites orbit)

From the polar equation of a conic section

$$r = \frac{p}{1 + e \cos v} \text{ Here } v \text{ represents true anomaly at new time}$$

In the perifocal coordinate system:

$$\hat{r} = \cos v \mathbf{P} + \sin v \mathbf{Q}$$

$$\hat{\theta} = -\sin v \mathbf{P} + \cos v \mathbf{Q}$$

We can write the position vectors in terms of its x and y components in perifocal system as

$$\mathbf{r} = r \cos v \mathbf{P} + r \sin v \mathbf{Q}$$

To obtain v we only need to differentiate r in equation keeping in mind that the perifocal coordinate frame is "inertial" and so $\dot{\mathbf{P}} = \dot{\mathbf{Q}} = 0$ and

Velocity in polar coordinates:

$$\mathbf{v} = \dot{r} \hat{r} + r \dot{v} \hat{\theta} = (\dot{r} \cos v - r \dot{v} \sin v) \mathbf{P} + (\dot{r} \sin v + r \dot{v} \cos v) \mathbf{Q}$$

This expression for $\dot{v}$ can be simplified by recognizing that $h = r^2 \dot{v}$, $p = \frac{h^2}{\mu}$ and differentiating polar equation of conics as

Radial velocity,

$$\dot{r} = \frac{dr}{dv} \frac{dv}{dt}$$

$$\frac{dr}{dv} = \frac{pe \sin v}{(1 + e \cos v)^2}$$

$$\dot{r} = \frac{pe \sin v}{(1 + e \cos v)^2} \dot{v}$$

From $h = r^2 \dot{v}$ and $h = \sqrt{\mu p}$

$$\dot{v} = \frac{h}{r^2} = \frac{\sqrt{\mu p}}{r^2}$$

$$\dot{r} = \frac{pe \sin v}{(1 + e \cos v)^2} \frac{\sqrt{\mu p}}{r^2}$$

But, $r = \frac{p}{1 + e \cos v}$ and $r^2 = \frac{p^2}{(1 + e \cos v)^2}$

Substitute for r^2

$$\dot{r} = \frac{pe \sin v}{(1 + e \cos v)^2} \frac{\sqrt{\mu p}}{p^2 / (1 + e \cos v)^2} = \frac{pe \sin v \sqrt{\mu p} (1 + e \cos v)^2}{p^2 (1 + e \cos v)^2}$$

$$\dot{r} = \frac{e \sin v \sqrt{\mu p}}{p}$$

$$\dot{r} = \sqrt{\frac{\mu}{p}} (e \sin v)$$

Transverse velocity, $r \dot{v}$

$$r \dot{v} = r \frac{h}{r^2} = \frac{h}{r} = \frac{\sqrt{\mu p}}{r} = \frac{\sqrt{\mu p}}{p / (1 + e \cos v)} = \sqrt{\frac{\mu}{p}} (1 + e \cos v)$$

$$r \dot{v} = \sqrt{\frac{\mu}{p}} (1 + e \cos v)$$

Substitute radial and transverse velocity in the polar equation

$$\mathbf{v} = (\dot{r} \cos v - r \dot{v} \sin v) \mathbf{P} + (\dot{r} \sin v + r \dot{v} \cos v) \mathbf{Q}$$

For $\mathbf{P}$:

$$e \sin v \cos v - (1 + e \cos v) \sin v = e \sin v \cos v - \sin v - e \cos v \sin v = -\sin v$$

For $\mathbf{Q}$:

$$e \sin^2 v + (1 + e \cos v) \cos v = e(\sin^2 v + \cos^2 v) + \cos v = e + \cos v$$

$$\mathbf{v} = \sqrt{\frac{\mu}{p}} [-\sin v \mathbf{P} + (e + \cos v) \mathbf{Q}]$$

Magnitude of velocity,

$$v = \sqrt{\dot{r}^2 + r\dot{v}^2} = \sqrt{\frac{\mu}{p}(e^2 \sin^2 v + (1 + e \cos v)^2)} = \sqrt{\frac{\mu}{p}(1 + 2e \cos v + e^2)} = \sqrt{\mu \left(\frac{2}{r} - \frac{1}{a} \right)}$$

This is the vis-viva equation obtained by polar equation of conics and

$$p = a(1 - e^2)$$

Transforming r and v from Perifocal system to Geocentric-equatorial-frame (Earth-Centred Inertial Frame)

The Perifocal Coordinate system is related geometrically to the IJK frame through the angles Ω , i , ω as shown in figure

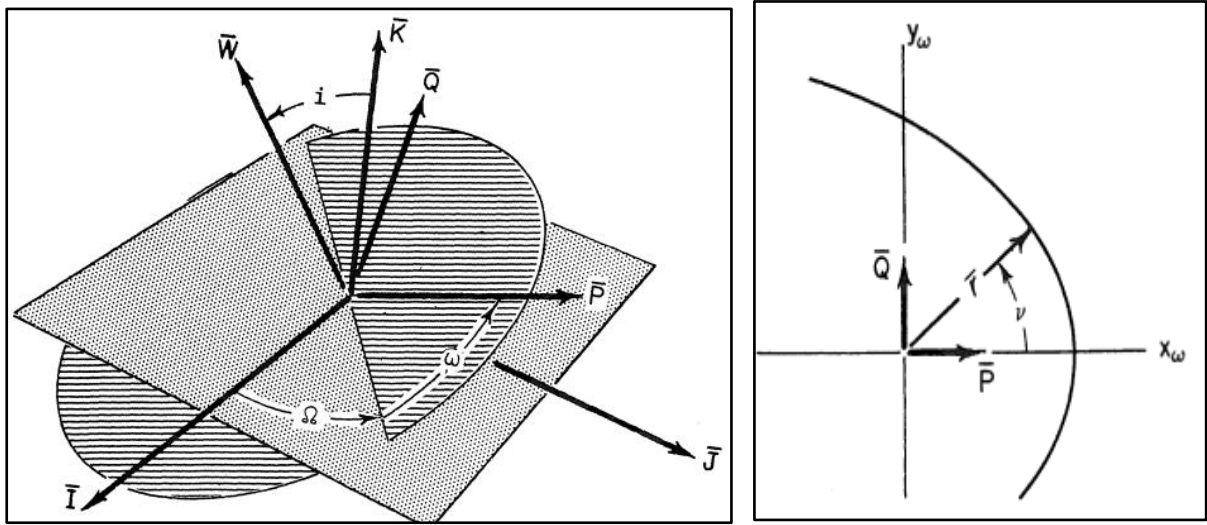


Figure 11: Perifocal Coordinate system

The transformation of coordinates can be achieved by means of a rotation matrix $\mathbf{R}$

Rotation about z-axis by ω

$$R_3(\omega) = \begin{bmatrix} \cos\omega & \sin\omega & 0 \\ -\sin\omega & \cos\omega & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Rotation about x-axis by i

$$R_1(i) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos i & \sin i \\ 0 & -\sin i & \cos i \end{bmatrix}$$

Rotation about z-axis by Ω

$$R_3(\Omega) = \begin{bmatrix} \cos\Omega & \sin\Omega & 0 \\ -\sin\Omega & \cos\Omega & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Compute, $R = R_3(\omega)R_1(i)R_3(\Omega)$

$$\mathbf{R} = \begin{bmatrix} I \cdot P & I \cdot Q & I \cdot W \\ J \cdot P & J \cdot Q & J \cdot W \\ K \cdot P & K \cdot Q & K \cdot W \end{bmatrix} = \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix}$$

The elements of $\mathbf{R}$ are

$$\begin{aligned}
R_{11} &= \cos\Omega\cos\omega - \sin\Omega\sin\omega\cos i \\
R_{12} &= -\cos\Omega\sin\omega - \sin\Omega\cos\omega\cos i \\
R_{13} &= \sin\Omega\sin i \\
R_{21} &= \sin\Omega\cos\omega + \cos\Omega\sin\omega\cos i \\
R_{22} &= -\sin\Omega\sin\omega + \cos\Omega\cos\omega\cos i \\
R_{23} &= -\cos\Omega\sin i \\
R_{31} &= \sin\omega\sin i \\
R_{32} &= \cos\omega\sin i \\
R_{33} &= \cos i
\end{aligned}$$

$$\begin{bmatrix} r_I \\ r_J \\ r_K \end{bmatrix} = \mathbf{R} \begin{bmatrix} r_P \\ r_Q \\ r_W \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} v_I \\ v_J \\ v_K \end{bmatrix} = \mathbf{R} \begin{bmatrix} v_P \\ v_Q \\ v_W \end{bmatrix}$$

Result 2: Computational Result of Ideal Keplerian Orbit in MATLAB (Appendix A)

The above derived equations are implemented in the MATLAB code of Appendix, which uses the initial position and velocity vectors of E08 satellite to determine the classical orbital elements. The orbital motion is then propagated in time by advancing the true anomaly through standard Keplerian relations. Based on the propagated orbital parameters, the satellite's position and velocity are evaluated in the perifocal reference frame. Finally, appropriate coordinate transformations are applied to convert the state vectors from the perifocal frame to the Earth-Centred Inertial frame. The results obtained are as follows:

Input State vectors

Input $\mathbf{r}_0 = [4771.810686 \ -3024.529533 \ 3804.492540]$ km

Input $\mathbf{v}_0 = [2.888529 \ 6.849725 \ 1.814147]$ km/s

Corresponding computed Orbital elements

$a = 6814.901875$ km
 $e = 0.000819$
 $i = 37.367910$ deg
 $RAAN = 265.771134$ deg
 $\omega = 115.120347$ deg
 $\nu_0 = 311.851501$ deg
 $\text{Period} = 5598.873197$ s (1.555243 hr)

Computed State vectors for 3 instances each after one complete revolution

v(deg)	t(s)	r_x(km)	r_y(km)	r_z(km)	v_x(km/s)	v_y(km/s)	v_z(km/s)
0	0.000000	4771.810686	-3024.529533	3804.492540	2.888529	6.849725	1.814147
360	5598.873197	4771.810686	-3024.529533	3804.492540	2.888529	6.849725	1.814147
720	11197.746393	4771.810686	-3024.529533	3804.492540	2.888529	6.849725	1.814147
1080	16796.619590	4771.810686	-3024.529533	3804.492540	2.888529	6.849725	1.814147

Plotting the position for each degree of true anomaly effectively constructs the entire orbital path.

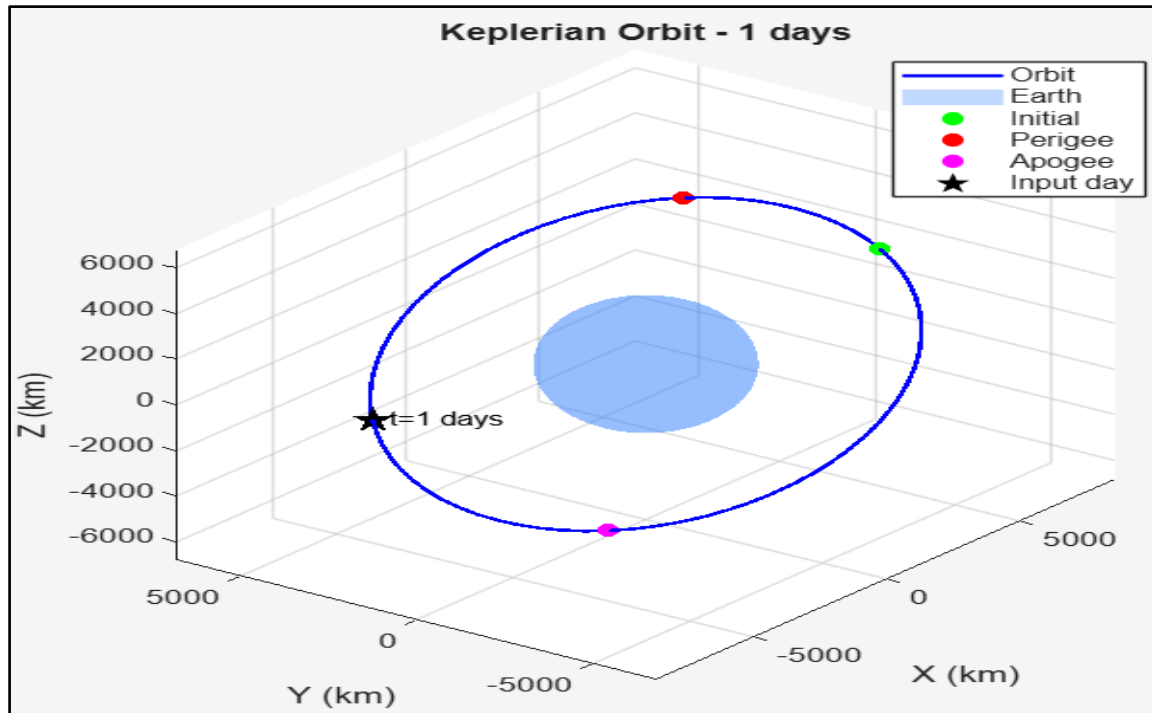


Figure 12: Ideal Keplerian Orbit

In a similar manner, when the time input is specified in days, the corresponding position and velocity vectors are determined using the same algorithm.

State Vector at $t = 1$ days:

$\mathbf{r}(t) = [-3282.072034 \quad 5279.297559 \quad -2796.894240] \text{ km}$

$\mathbf{v}(t) = [-4.849545 \quad -4.820077 \quad -3.421937] \text{ km/s}$

Since this is an ideal Keplerian orbit, the satellite's position will always remain on the same fixed trajectory. This behaviour aligns with the intuitive assumption one might make before studying orbital mechanics—that a satellite continuously follows a single, unchanging path.

The following section investigates the perturbing forces acting on the satellite and quantifies their impact on the orbital trajectory (especially J2 and Atmospheric Drag).

Orbital Perturbations

A perturbation is defined as the separation between a satellite's real orbit and its ideal Keplerian orbit.

According to Kepler's first law, a satellite's orbit is described using Keplerian elements, assuming Earth is a perfect sphere and concentrating its mass at the centre. Under this condition, the centrifugal force balances the gravitational pull. However, in reality, several additional forces—known as perturbing forces—act on the satellite, driving it away from the ideal Keplerian orbit. These perturbations must be accounted for when predicting satellite position and planning orbit maintenance.

Once placed in orbit, a satellite experiences various perturbing torques that alter its orbital parameters over time.

The main sources include:

- The asymmetry of Earth's gravitational field
- Atmospheric drag
- Gravitational attractions from the Sun, Moon, and other celestial bodies
- Solar radiation pressure
- Magnetic field interactions
- Earth tides

These perturbations cause orbital drift and orientation variations. Therefore, orbit control is necessary to maintain the satellite's correct position, both east–west and north–south. For example, a 1° drift in longitude for a geostationary satellite corresponds to an approximate displacement of 735 km along its orbit.

These perturbations can be broadly classified as gravitational and non-gravitational forces.

Gravitational Perturbations:

- Earth's oblateness (non-sphericity)
- Third-body attractions from the Sun and Moon

Non-Gravitational Perturbations:

1. Atmospheric drag
2. Solar radiation pressure
3. Outgassing (fuel leaks from spacecraft tanks)
4. Tidal friction effects

These can be further grouped as conservative and nonconservative perturbations:

- Conservative forces depend only on position and involve no energy exchange. The total orbital energy remains constant, so the semi-major axis does not change. Examples: central-body and third-body gravitational effects.
- Nonconservative forces depend on both position and velocity, leading to energy gain or loss (e.g., via friction or thrust). Examples: atmospheric drag, tidal friction, and solar radiation pressure.

Keplerian motion is based on the two-body problem, which assumes each body (Earth and satellite) acts as a point mass with spherical symmetry. The real-world case introduces additional gravitational and non-gravitational forces, making it a three-body or n-body problem.

In the perturbed case, the actual motion can no longer be expressed exactly using simple conic equations (ellipse, hyperbola, or parabola). The effects of various perturbing accelerations reduce the accuracy of the two-body approximation over time.

The general equation of motion including perturbations is:

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r} + \mathbf{a}_p$$

where $\mathbf{a}_p$ represents the sum of all perturbing accelerations.

Thus, to determine the state vectors considering the perturbations, we need to consider the same algorithm used in Ideal Keplerian orbit and apply the changes due to Perturbations.

Causes and Effects of Perturbations

1. Small continuous forces in space can cause acceleration over time, moving the satellite away from its target position.
2. Earth's nonspherical shape (flattened poles and equatorial bulge) causes the gravitational field to vary with latitude and longitude.
 - Earth's equatorial diameter exceeds the polar diameter by about 20 km.
 - Density variations within Earth create non-uniform gravitational pulls.
3. Gravitational influences from the Sun and Moon change direction over daily, monthly, and yearly cycles.
4. Solar activity such as solar winds and flares can affect motion and disrupt satellite communication.

The calculation of orbital perturbations is critically important in satellite operations as it helps determine the satellite's operational lifetime, plan station-keeping manoeuvres, and manage fuel usage effectively. Understanding these perturbations allows operators to anticipate and correct deviations in a satellite's path caused by gravitational and non-gravitational forces. This is essential for maintaining the satellite's orientation and positional stability, which in turn prevents communication interference with neighbouring satellites and ensures accurate coverage of designated regions on Earth. Therefore, sustaining a functional and efficient orbit requires continuous monitoring and periodic corrections to counteract the combined effects of all perturbing influences, ensuring long-term reliability and mission success.

1. Perturbations due to flat poles (J2 Effect)

Earth isn't a perfect sphere—it's slightly flattened at the poles and bulging at the equator due to its rotation, making it an oblate spheroid. The equatorial diameter is roughly 20–40 km larger than the polar diameter, and the Earth's density varies in different regions, causing its gravity field to be uneven.

For satellites orbiting Earth, this uneven gravity from the Earth's oblateness causes their orbits to change over time. The bulge creates a force pulling satellites back toward the equatorial plane if they drift above or below it. Additionally, the orbit's orientation slowly shifts, shown by changes in the right ascension of the ascending node and the argument of perigee—basically, the orbit slowly rotates relative to Earth.

Mathematically, Earth's uneven gravity field is described using gravity harmonics, with J2 being the largest term. J2 is directly related to Earth's equatorial bulge and causes the major long-term changes in satellite orbits, including gradual drifting and rotation of the orbital plane. For a spherical earth of uniform mass, Kepler's third law gives the nominal mean motion n_0 as

$$n_0 = \sqrt{\frac{\mu}{a^3}}$$

The 0 subscript is included as a reminder that this result applies for a perfectly spherical earth of uniform mass. However, it is known that the earth is not perfectly spherical, there being an equatorial bulge and a flattening at the poles, a shape described as an oblate spheroid. When the earth's oblateness is considered, the mean motion, denoted in this case by symbol n , is modified to

$$n = n_0 \left[1 + \frac{K_1(1 - 1.5 \sin^2 i)}{a^2(1 - e^2)^{3/2}} \right]$$

Where,

$$K_1 = \frac{3}{2} J_2 R_e^2 = 66,063.1704 \text{ km}^2$$

The new mean motion, thereby, is

$$\bar{n} = \sqrt{\frac{\mu}{a_0^3}} \left[1 + \frac{3 J_2 R^2}{2 p^2} \left(1 - \frac{3}{2} \sin^2 i \right) (1 - e^2)^{\frac{1}{2}} \right]$$

The orbital period considering the earth's oblateness is termed the anomalistic period (e.g., from perigee to perigee). The mean motion is the reciprocal of the anomalistic period. The anomalistic period is

$$P_A = \frac{2\pi}{n} s$$

where n is in radians per second.

The oblateness of the earth also produces two rotations of the orbital plane. The first of these, known as regression of the nodes, is where the nodes appear to slide along the equator. In effect,

the line of nodes, which is in the equatorial plane, rotates about the centre of the earth. Thus Ω , the right ascension of the ascending node, shifts its position.

If the orbit is prograde, the nodes slide westward, and if retrograde, they slide eastward. As seen from the ascending node, a satellite in prograde orbit moves eastward, and in a retrograde orbit, westward. The nodes therefore move in a direction opposite to the direction of satellite motion, hence the term regression of the nodes. For a polar orbit ($i=90^\circ$), the regression is zero.

Both effects depend on the mean motion n , the semimajor axis a , and the eccentricity e . These factors can be grouped into one factor K given by

$$K = \frac{nK_1}{a^2(1 - e^2)^2}$$

An expression for the rate of change of Ω with respect to time is

$$\frac{d\Omega}{dt} = -K \cos i$$

$$\dot{\Omega} = -\frac{3J_2R^2}{2p^2}\bar{n}\cos i$$

When the inclination i is equal to 90° , cosine angle is equal to zero, so that the regression of the nodes is zero, and the rate of rotation of the line of apsides is given by $-K/2$.

When the rate of change is negative, the regression is westward, and when the rate is positive, the regression is east ward. Therefore, for eastward regression, i must be greater than 90° , or the orbit must be retrograde.

The other major effect produced by the equatorial bulge is a rotation of the line of apsides. This line rotates in the orbital plane, resulting in the argument of perigee changing with time.

Due to the perturbed orbit, the satellite does not return to the same point in space after one revolution. The time elapsed between the successive perigee passages is referred to as anomalistic period. The anomalistic period (T_A) is given by equation.

$$T_A = \frac{2\pi}{\omega}$$

Where,

$$\omega = \omega_0 \left[1 + \frac{K(1 - 1.5 \sin^2 i)}{a^2(1 - e^2)^{3/2}} \right]$$

ω_0 is the angular velocity for spherical Earth or simply the mean motion n ,

$K = 66063.1704 \text{ km}^2$, a is the semi-major axis, e is the eccentricity and i is inclination.

The rate of change is given by

$$\frac{d\omega}{dt} = K(2 - 2.5 \sin^2 i)$$

$$\dot{\omega} = \frac{3J_2R^2}{2p^2} \bar{n} \left(2 - \frac{5}{2} \sin^2 i \right)$$

When the inclination i is equal to 63.435° , the term within the parentheses is equal to zero, and hence no rotation takes place.

Denoting the epoch time by t_0 , the right ascension of the ascending node by Ω_0 , and the argument of perigee by ω_0 at epoch gives the new values for Ω and ω at time t as

$$\Omega = \Omega_0 + \frac{d\Omega}{dt} (t - t_0)$$

$$\omega = \omega_0 + \frac{d\omega}{dt} (t - t_0)$$

Where,

- $\dot{\Omega}$ = rate of ascending node
- $\dot{\omega}$ = argument of perigee rate
- $\bar{n}$ = orbit mean motion with J_2 correction
- $J_2 = 0.00108263$
- R = Earth equatorial radius
- i = orbit inclination
- μ = gravitational constant
- a_0 = semimajor axis at epoch
- e = eccentricity
- $p = a_0(1 - e^2)$

Result 3: Computational Result considering J2 Perturbations (Appendix B)

The equation of obtaining the new mean motion due to J_2 effect, is used in determining the new RAAN and ω over time, these perturbed orbital elements are then implemented in the same algorithm as the Ideal Keplerian orbit. The computed results in MATLAB code shown in Appendix B are as follows:

Initial State Vectors of E08:

Input $r_0 = [4771.810686 \ -3024.529533 \ 3804.492540]$ km

Input $v_0 = [2.888529 \ 6.849725 \ 1.814147]$ km/s

Change in orbital period

$T_0 = 5598.873197$ s, $T_{J2} = 5595.311874$ s

The rate of change of Ω and ω

J2 rates: $d\text{RAAN}/dt = -1.269489\text{e-}06$ rad/s (-6.284422 deg/day)

$d\omega/dt = 1.723664\text{e-}06$ rad/s (8.532747 deg/day)

State at $t = 2$ days (from r_0, v_0 epoch):

$r = [1996.178973, -5901.742932, 2761.867997]$ km

$v = [5.805478, 3.586648, 3.452743]$ km/s

RAAN = 253.202291 deg

$\omega = 132.185841$ deg

$\nu = 269.706466$ deg

$a = 6814.901875$ km

$e = 0.000819$

$i = 37.367910$ deg

$M = 269.800299$ deg

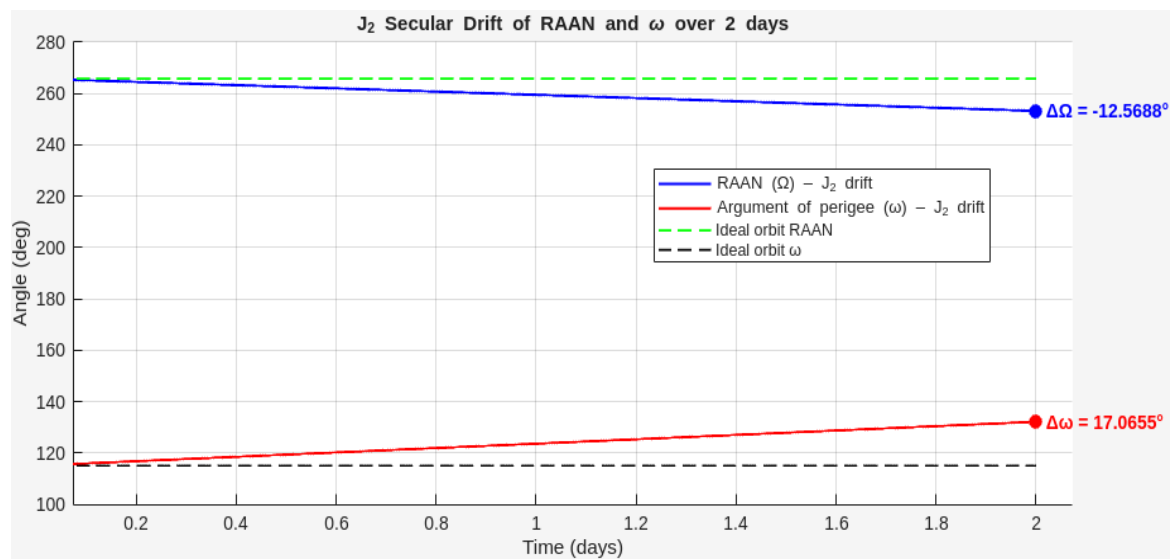


Figure 13: Rate of change in RAAN and ω of E08

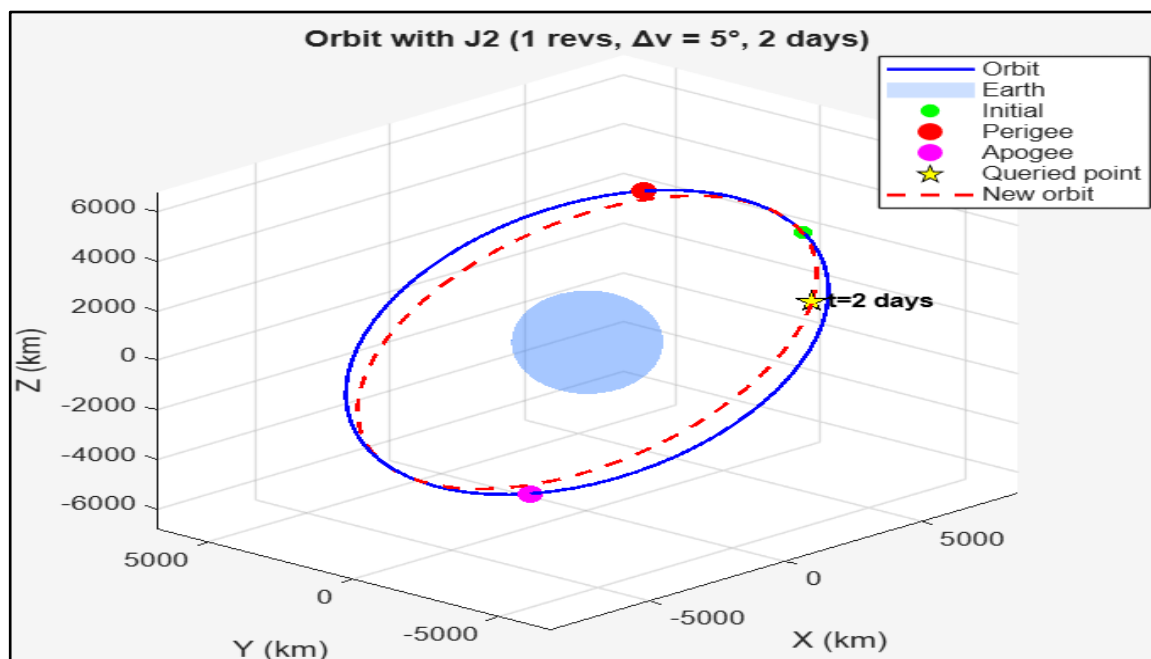


Figure 14: E08 Orbit with J2

Plots of E06 Satellite

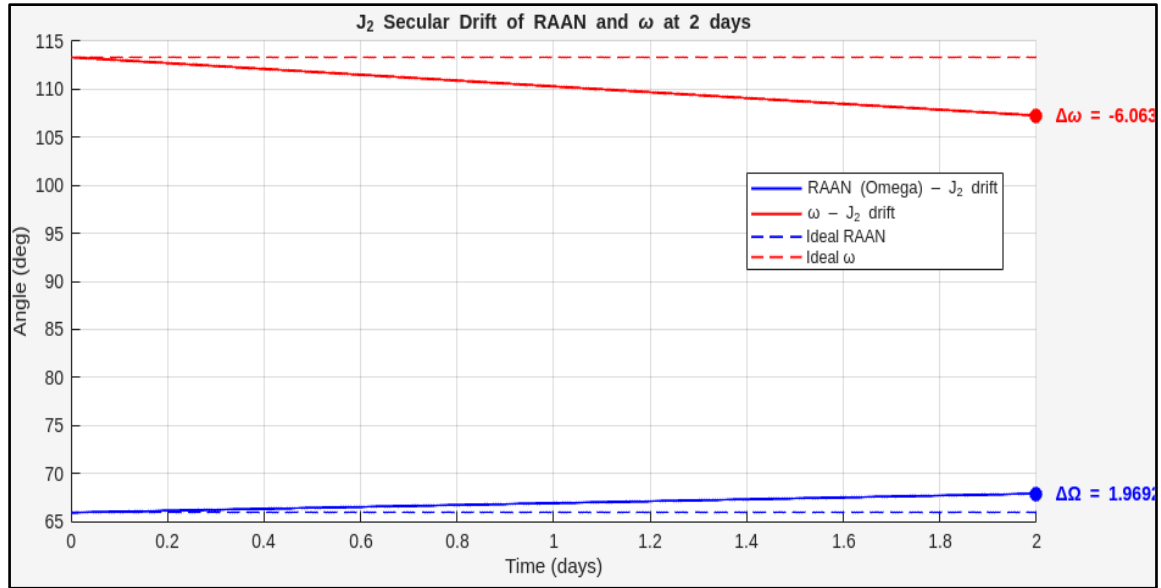


Figure 15: E06 J2 drift

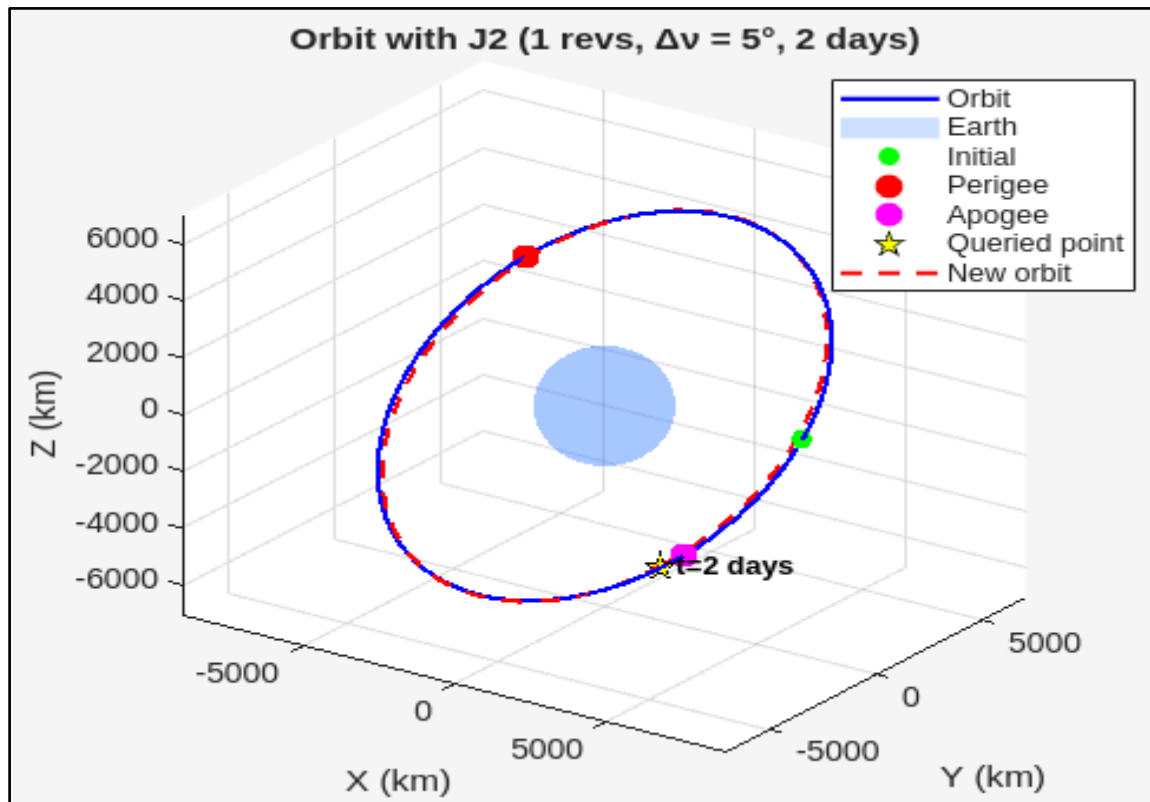


Figure 16: E06 orbit with J2

The above figures illustrate the changes in the satellite's orbit caused by the J₂ perturbation. Below are the data of two satellites, E08 and E06. E08 is present at about 450km altitude, and E06 is at 740km altitude. Thereby comparing a satellite which is highly affected by the perturbations (E08) and another satellite which is less significantly affected by the perturbations (E06)

E08 With J2													
After 1 day	r_x	r_y	r_z	v_x	v_y	v_z	a	e	i	w	O	v	M
Computed	-3621.612006	4688.260513	-3372.561431	-4.600976	-5.481387	-2.690858	6814.901875	0.000819	37.36791	123.653094	259.486713	110.948484	110.860833
STK Value	-3622.483434	4687.202475	-3373.06884	-4.600114	-5.482528	-2.690051	6814.894313	0.000818	37.368	123.711	259.487	110.903	110.815
Difference	0.871428	1.058038	0.507409	-0.000862	0.001141	-0.000807	0.007562	1E-06	-9E-05	-0.057906	-0.000287	0.045484	0.045833
After 2 day	r_x	r_y	r_z	v_x	v_y	v_z	a	e	i	w	O	v	M
Computed	1996.178973	-5901.742932	2761.867997	5.805478	3.586648	3.452743	6814.901875	0.000819	37.36791	132.185841	253.202291	269.706466	269.800299
STK Value	1998.413067	-5900.359418	2763.197327	5.804514	3.589505	3.451406	6814.894313	0.000818	37.368	132.244	253.202	269.673	269.767
Difference	-2.234094	-1.383514	-1.32933	0.000964	-0.002857	0.001337	0.007562	1E-06	-9E-05	-0.058159	0.000291	0.033466	0.033299

E06 With J2													
After 1 day	r_x	r_y	r_z	v_x	v_y	v_z	a	e	i	w	O	v	M
Computed	-1190.864307	-4732.766352	5174.622695	-2.837351	-4.778687	-5.019988	7119.04121	0.000926	98.35213	110.295017	66.920831	22.373053	22.332679
STK Value	-1191.37243	-4733.616684	5173.709042	-2.837118	-4.777752	-5.021032	7119.033913	0.000927	98.352	110.337	66.921	22.341	22.301
Difference	0.508123	0.850332	0.913653	-0.000233	-0.000935	0.001044	0.007297	-1E-06	0.00013	-0.041983	-0.000169	0.032053	0.031679
After 7 day	r_x	r_y	r_z	v_x	v_y	v_z	a	e	i	w	O	v	M
Computed	2000.42421	6801.267986	-658.731232	1.238411	0.342436	7.370643	7119.04121	0.000926	98.35213	92.105249	72.828427	262.529091	262.634338
STK Value	2001.984885	6801.693047	-649.42254	1.235619	0.332946	7.371557	7119.033913	0.000927	98.352	92.148	72.828	262.563	262.668
Difference	-1.560675	-0.425061	-9.308692	0.002792	0.00949	-0.000914	0.007297	-1E-06	0.00013	-0.042751	0.000427	-0.033909	-0.033662

The computed data is compared with the value of Ansys STK computed data which has inbuilt J2 model

For both E06 and E08 cases, the **J₂-only results** show very small differences between the computed and STK computed state vectors. The variations in position, velocity, and orbital elements remain consistently low, confirming that the J₂ model successfully captures the dominant geopotential effect with high accuracy.

2. Perturbations due to Atmospheric Drag

Satellites orbiting a few hundred kilometres above Earth's surface experience an additional force caused by atmospheric resistance, also known as drag. Because the atmosphere becomes thinner exponentially with height, this effect is most significant for low-Earth orbit (LEO) satellites and is strongest when the satellite passes through the lowest point of its orbit, known as the perigee.

This atmospheric drag reduces the satellite's kinetic energy and angular momentum, which gradually lowers both the semi-major axis and the orbital eccentricity. For satellites without active control systems, this slow loss of altitude and the tendency of the orbit to become more circular severely limit their operational lifetimes. However, since drag acts mainly in the direction opposite to the satellite's motion, it has little to no effect on the orientation of the orbital plane.

Unlike gravitational forces—which act equally regardless of a satellite's size or mass—drag and other surface forces depend on the satellite's physical characteristics, such as mass, shape, and surface area.

When a satellite's perigee drops below about 1,000 kilometres, the effects of atmospheric drag become increasingly important. Because drag is a non-conservative force, it continuously removes energy from the orbit. As a result, the semi-major axis and the orbital period gradually decrease.

Interestingly, even though the orbit is shrinking, the satellite's velocity increases in accordance with Kepler's law, expressed as

$$\mu = n^2 a^3 = \text{constant.}$$

This phenomenon is known as the drag paradox—friction slows a satellite down in terms of energy, yet it moves faster as it spirals inward toward Earth. Since drag is greatest at perigee—where both velocity and atmospheric density are highest—the largest energy loss occurs there. This causes the orbit to become increasingly circular over time. Eventually, the satellite's apogee lowers until it matches the perigee altitude, producing a circular orbit. From that stage, the satellite rapidly descends into denser layers of the atmosphere and ultimately re-enters.

For near-circular orbits, the orbit semimajor axis decay rate can be computed by the following simple equation:

$$\frac{da}{dt} = -na^2 \rho B$$

Where,

n = mean motion

ρ = atmosphere density at that altitude

B = ballistic coefficient = $C_d A/m$

m = total mass of spacecraft, kg

A = effective cross-section area, m^2

C_d = drag coefficient = 2.0

Accurately predicting a satellite's motion under the influence of atmospheric drag requires a reliable model of the upper atmosphere's density. Over the past several decades, scientists have developed a range of such models, each offering different levels of detail and precision.

The NRLMSISE-00 model is one of the most advanced and widely used atmospheric density models. It accounts for a broad range of physical influences on the upper atmosphere, including variations driven by solar extreme ultraviolet radiation, geomagnetic activity, seasonal and latitudinal changes, and day–night differences. In addition, it incorporates long-term solar cycle effects, short-term geomagnetic disturbances, and empirical data from satellite observations, making it highly accurate for predicting density, temperature, and atmospheric composition from the ground up to the exosphere.

***NRLMSISE-00* uses the following laws for determining density**

(A) Hydrostatic and Ideal Gas Law

(Hedin 1987; MSIS-86)

Hydrostatic:

$$\frac{dP}{dz} = -\rho g$$

This says the pressure decreases as you go higher because the air above pushes down with gravity (weight of air).

Ideal Gas Law:

$$P = nk_B T$$

This relates pressure P to the number of particles n and temperature T , saying warmer air or more particles mean higher pressure.

Combined Number Density Equation:

$$\frac{1}{n} \frac{dn}{dz} = -\frac{mg}{k_B T}$$

Integrated (non-isothermal):

$$n(z) = n(z_0) \left(\frac{T_0}{T(z)} \right)^{1+\alpha_i} \exp \left(- \int_{z_0}^z \frac{m_i g(z')}{k_B T(z')} dz' \right)$$

These are fundamental physics laws describing how air pressure and temperature are linked to density and altitude. The model uses them to find how particle density (n) changes with height.

(B) Bates Temperature Profile

(Bates 1959)

$$T(z) = T_\infty - [T_\infty - T_0] \exp [-S(z - z_0)]$$

Where:

T_∞ : Exospheric temperature

T_0 : Temperature at base altitude (z_0 , typically 120 km)

S : Inverse scale length ($\approx 0.018 \text{ km}^{-1}$)

The temperature gradually changes from a base value at about 120 km (T_0) to a nearly constant value in the upper atmosphere called the exospheric temperature (T_∞).

The temperature in the upper atmosphere doesn't stay constant; it rises and approaches a maximum. This formula models that smooth rise.

(C) Diffusive Equilibrium

(Walker 1965)

Differential Form:

$$\frac{1}{n_i} \frac{dn_i}{dz} = -\frac{1}{T} \left(\frac{m_i g}{k_B} + \alpha_i \frac{dT}{dz} \right)$$

Integrated Solution:

$$n_i(z) = n_i(z_0) \left(\frac{T_0}{T(z)} \right)^{1+\alpha_i} \exp \left[- \int_{z_0}^z \frac{m_i g(z')}{k_B T(z')} dz' \right]$$

Different kinds of gas particles (like oxygen or helium) spread out differently in the atmosphere because they have different masses. This formula shows how their density changes with height and temperature.

To get accurate air composition at each altitude since lighter gases stay higher and heavier gases stay lower.

(D) Total Mass Density and Pressure

(Picone et al., 2002)

Total Mass Density:

$$\rho(z) = \sum_i n_i(z) m_i$$

Pressure:

$$P(z) = \sum_i n_i(z) k_B T(z)$$

This sums up the mass and pressure contributions from all types of gas particles.

(E) Anomalous Oxygen Component

(Picone et al., 2002)

$$n_{AO}(z) = n_{AO,0} \exp \left[-\frac{(z - z_{AO})}{H_{AO}} \right]$$

Where: $H_{AO} \approx 200\text{km}$; used above $\approx 500\text{ km}$

This represents a special population of oxygen atoms that behave differently above roughly 500 km altitude.

(F) Solar EUV / Solar Flux Effect on Exospheric Temperature

(Picone et al., 2002)

$$T_{\infty} = T_0 + C_1 \cdot (F_{10.7} - 150) + C_2 \cdot (F_{10.7A} - 150) + C_3 \cdot Ap + \dots$$

The exospheric temperature T_{∞} (temperature far above 120 km) depends on solar activity and geomagnetic storms. The model takes the baseline temperature T_0 and adjusts it by how strong the solar radiation is right now and recently ($F_{10.7}$ daily and 81-day average) and geomagnetic activity (Ap index).

(G) Geomagnetic Heating Corrections

(Picone et al., 2002)

$$\Delta T(z) = \sum_j D_j \cdot f_j(z) \cdot Ap_j$$

This describes how geomagnetic storms inject energy into the atmosphere at different heights z . The coefficients D_j and functions $f_j(z)$ determine how much extra heating occurs at each altitude for various geomagnetic indices Ap_j .

(H) Species-Specific Density Corrections

(Picone et al., 2002)

$$n'_i(z) = n_i(z) \cdot \left(1 + \sum_k E_k \cdot g_k(z) \right)$$

This formula applies corrections to the densities of each gas species i , accounting for complex physical and chemical processes not covered by simple diffusion. E_k are correction factors, and $g_k(z)$ are altitude-dependent functions.

(I) Seasonal and Diurnal Variations of Temperature

(Picone et al., 2002)

$$T(z, t, \phi, \lambda) = T(z) + A_1 \cos\left(\frac{2\pi t}{24} - \phi\right) + A_2 \cos\left(\frac{4\pi t}{24} - 2\phi\right) + \dots$$

This equation models how temperature changes with time of day t , latitude ϕ , and longitude λ , capturing daily (diurnal) and twice-daily (semidiurnal) temperature cycles with amplitudes A_1, A_2 . The Sun heats different parts of the atmosphere depending on time of day, causing temperature and density to rise and fall in patterns. This formula helps the model recreate those regular oscillations that satellites experience.

(J) Polynomial or Spline Fits for Vertical Profiles

$$Q(z) = \sum_{m=0}^M a_m (z - z_r)^m$$

This is a mathematical way to smoothly describe how any parameter Q (such as temperature, density corrections, or other profile values) changes with altitude z , using coefficients a_m and a reference altitude z_r .

Influence of Each Input Parameter

Input	What It Is / How Measured	Effect on Temperature	Effect on Density	Explanation
Solar Flux (F10.7, F10.7A)	Measurement of solar radio emission at 10.7 cm wavelength. F10.7 is daily solar flux; F10.7A is 81-day smoothed average.	Higher solar flux means more solar energy absorbed, which increases the temperature of the upper atmosphere (exosphere).	Higher temperatures cause the atmosphere to expand, increasing the scale height, so density decreases less rapidly with altitude.	Imagine the Sun heating the upper atmosphere; more solar energy makes it hotter and 'puffier,' affecting how thin or thick the air is at different heights.
Geomagnetic Index (Ap)	A measure of geomagnetic storm activity on Earth, changes hourly and daily.	Geomagnetic storms inject energy that heats the upper atmosphere temporarily, raising temperatures.	This heating causes the atmosphere to expand, temporarily increasing density in the thermosphere and affecting satellite drag.	Geomagnetic storms work like a sudden heat blast, making the air hotter and more expanded, leading to a denser atmosphere at high altitudes for a short time.

Input	What It Is / How Measured	Effect on Temperature	Effect on Density	Explanation
Altitude (z)	The height above Earth's surface (in km).	Temperature varies with altitude following a profile; generally, temperature decreases then increases with height per thermospheric physics.	Density decreases roughly exponentially with altitude because the air thins rapidly the higher you go.	The higher you go, the colder or hotter the air is depending on altitude, but the air always gets thinner, so density drops sharply with height.
Latitude (ϕ)	Position north or south of the equator (degrees).	Affects temperature by controlling how much sunlight an area receives seasonally and geomagnetic heating (more at poles).	Density varies with latitude due to solar heating patterns and geomagnetic effects altering atmospheric composition and expansion.	Near poles, auroras heat the atmosphere, and at equator, solar heating is stronger during day; this changes local temperature and density.
Longitude (λ)	Position east or west around the Earth (degrees).	Minor effect, mostly affecting timing of solar heating (local solar time offsets).	Causes slight variations in density because of geographic and magnetic field variations.	Longitude shifts when daytime occurs; this timing difference slightly changes local heating and atmosphere properties.
Local Solar Time (t_{lst})	Time of day at a given longitude (hours).	Introduces daily temperature cycles (warmer days, cooler nights), including tides in atmosphere.	Causes density to vary by up to $\pm 20\%$ during the day-night cycle due to warming and cooling cycles.	Like the weather changing from day to night, temperature and density cycle daily, with denser air in the day due to heating.

Input	What It Is / How Measured	Effect on Temperature	Effect on Density	Explanation
Day of Year (DOY)	Calendar day number (1 to 365/366), tracking seasons.	Seasonal changes impact temperature through varying solar zenith angles and sun intensity.	Density changes seasonally, thicker in summer with more solar heating, thinner in winter.	The Earth's tilt causes seasons, changing sunlight amount and angle, which warms or cools the atmosphere differently over the year

Result 4: Computational Result considering both J2 and Drag (Appendix C & D)

Using the inbuilt MATLAB function **atmosnrlmsise00**, the atmospheric density at the satellite's altitude is obtained. This density value is then applied in the drag equation to compute the resulting decay in the semi-major axis.

Input State Vectors of E08:

Input $r_0 = [4771.810686 \ -3024.529533 \ 3804.492540]$ km

Input $v_0 = [2.888529 \ 6.849725 \ 1.814147]$ km/s

Corresponding computed orbital elements

Elements from r_0, v_0 : $a=6814.901875$ km, $e=0.000819$, $i=37.367910$ deg

RAAN= 265.771134 deg, $\omega=115.120347$ deg, $\nu_0=311.851501$ deg

$T_0 = 5598.873197$ s

Computed Decayed semi major axis

For 1 revolution

Latitude = 33.956696762886807°

Longitude = $-79.152878846426105^\circ$

Altitude = 433.039780945835446 km

Temperature = 874.881869782696299 K

Density = $1.009012595156535e-12$ kg/m³

$da/dt = -7.248387549424036e-04$ m/s

a (initial) = 6814.901875488068981 km

a (decayed) = 6814.892835860317973 km

The decayed semi-major axis is substituted into the J_2 perturbation model, allowing computation of the satellite's state vector with the combined effects of oblateness and atmospheric drag.

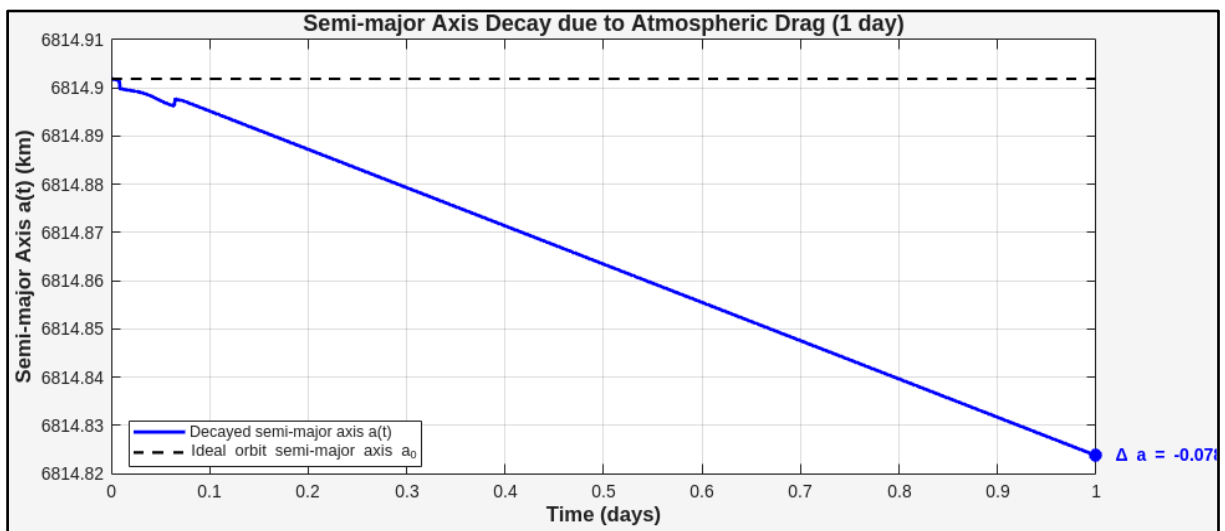


Figure 17: Semi Major axis decay of E08

Queried state vector at 1 days using decayed a_{final} :

$\mathbf{r} = [-3628.411947, 4680.043162, -3376.522859]$ km

$\mathbf{v} = [-4.594215, -5.490189, -2.684554]$ km/s

$\mathbf{v} = 111.044106$ deg

$a = 6814.823746$ km

$e = 0.000819$

$i = 37.367910$ deg

$\omega = 123.653113$ deg

RAAN = 259.486699 deg

$M = 110.956511$

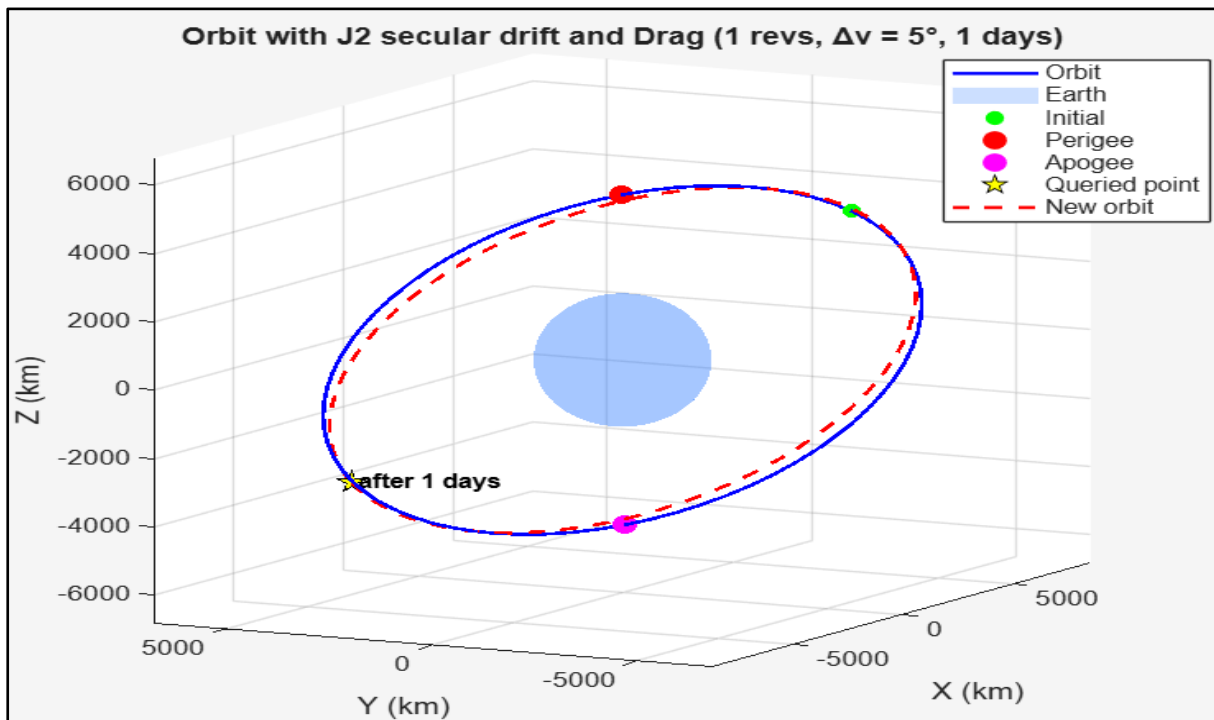


Figure 18: E08 Orbit with J2 and Drag

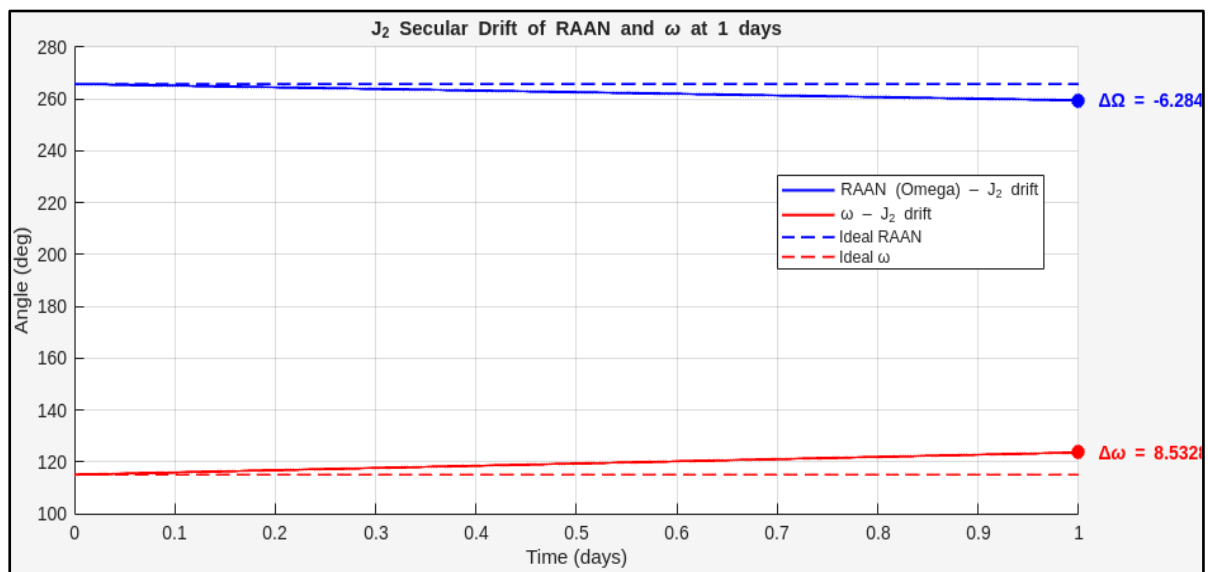


Figure 19: Rate of change of RAAN and ω of E08

Plot for E06 satellite:

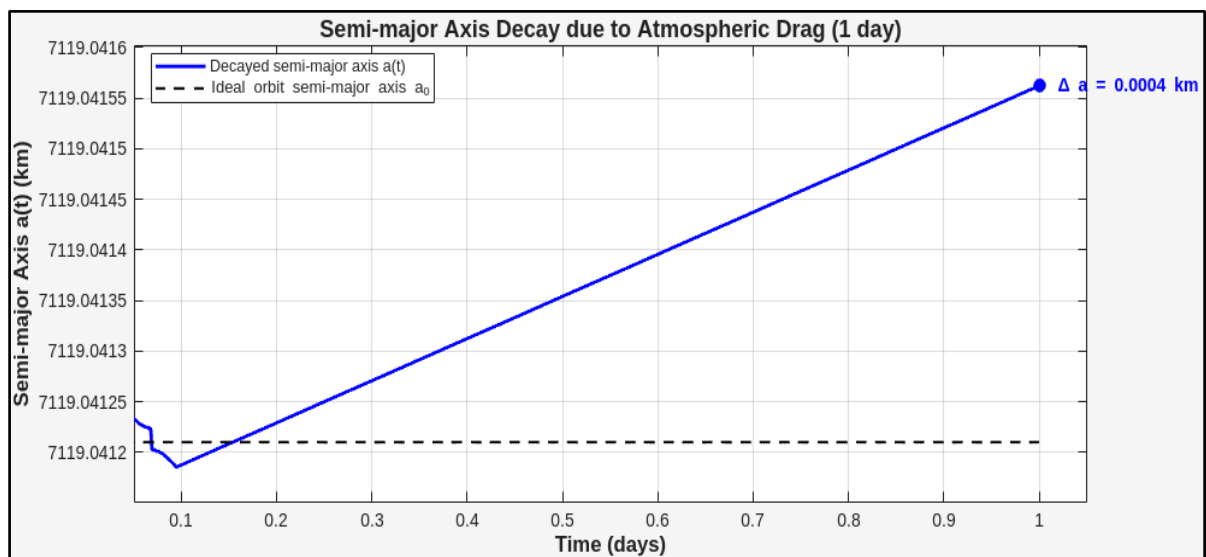


Figure 20: Semi major axis decay for E06

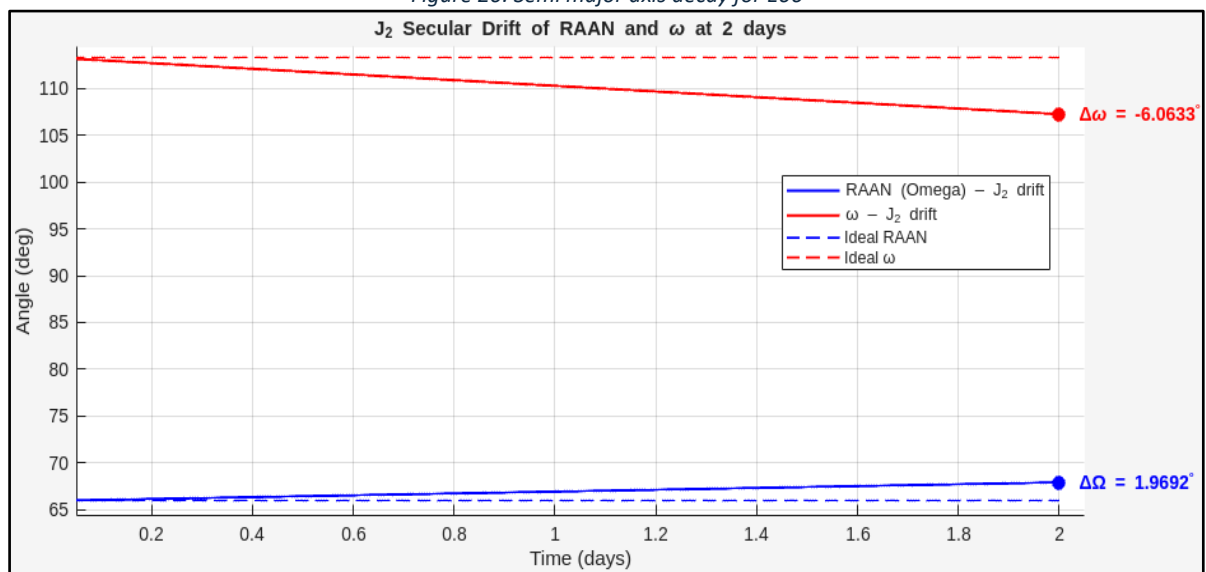


Figure 21: E06 J2 drift

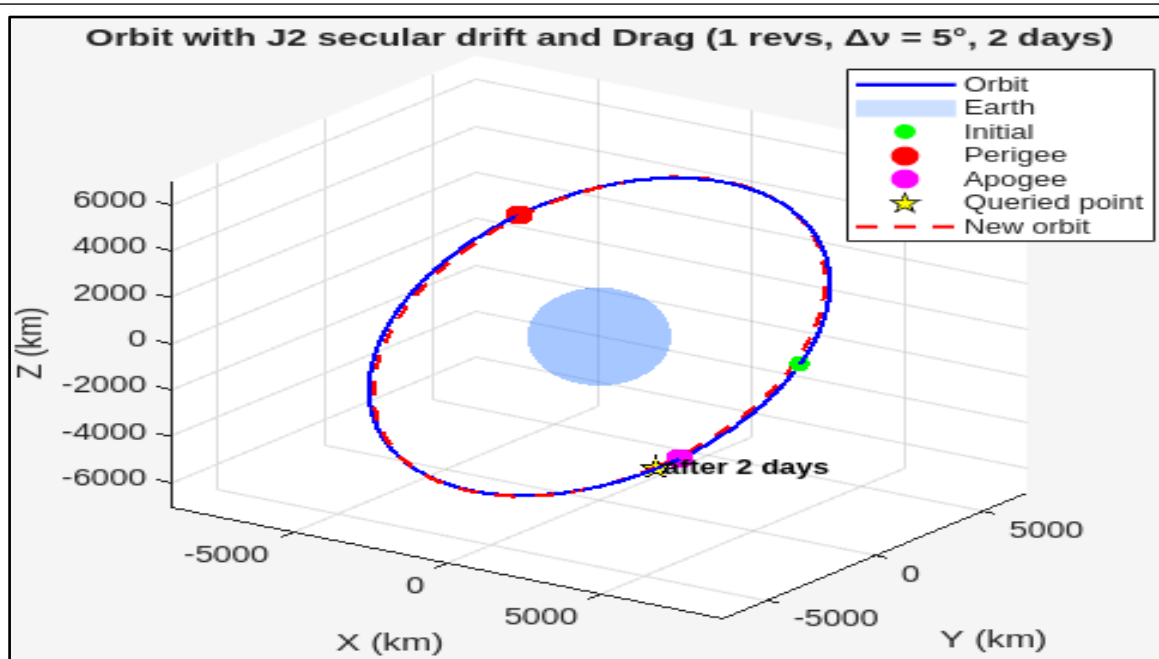


Figure 22: E06 orbit with J2+drag

Anslys STK Software doesn't include any in-built model which only include J2 and Drag, instead it includes every Earth Gravity Harmonics such as Zonal (J3, J4, J22), Tesseral and Sectoral.

E08 With J2 and Drag													
After 1 day	r_x	r_y	r_z	v_x	v_y	v_z	a	e	i	w	O	v	M
Computed	1548.233239	-6157.494708	2490.914519	5.972296	3.001575	3.705759	6816.470031	0.000569	37.377478	207.395627	253.212927	189.590311	189.601118
STK Value	1443.466312	-6201.395372	2425.834973	6.010019	2.875505	3.761941	6818.365515	0.000915	37.389	76.543	253.201	319.354	319.422
Difference	104.766927	43.900664	65.079546	-0.037723	0.12607	-0.056182	-1.895484	-0.000346	-0.011522	130.852627	0.011927	-129.763689	-129.82082

E06 With J2 and Drag													
After 1 day	r_x	r_y	r_z	v_x	v_y	v_z	a	e	i	w	O	v	M
Computed	-1629.567966	-5434.922416	4295.025672	-2.583287	-3.867533	-5.86506	7118.485015	0.000793	98.352249	75.615726	66.928787	66.793234	66.709755
STK Value	-1976.934501	-5919.753815	3409.022422	-2.299989	-2.973195	-6.477016	7115.459215	0.00138	98.354	84.829	66.932	66.191	66.046
Difference	347.366535	484.831399	886.00325	-0.283298	-0.894338	0.611956	3.0258	-0.000587	-0.001751	-9.213274	-0.003213	0.602234	0.663755

Thus, the actual data mentioned in the table is obtained for all the perturbations of Gravity Harmonics. Thereby, the calculated difference is expected to be more significant and will not match with the data from STK.

In general, once atmospheric drag is included, the behaviour changes noticeably. Drag introduces a gradual **reduction in the semi-major axis**, which in turn results in larger deviations in the satellite's computed position, velocity, and orbital elements especially over multi-day propagation. The differences in position and velocity components become larger compared to the J₂-only case, showing that the orbit decays and the satellite move faster along a slightly lower trajectory.

This trend is particularly visible in the argument of perigee, RAAN, and true anomaly, which show increased sensitivity once drag is applied. The E08 case, which is at a lower orbit compared to E06, exhibits even stronger variations matching the expectation that drag effects grow rapidly as altitude decreases.

Other Small Perturbations in LEO

1. Solar Radiation Pressure Effects

Solar radiation pressure doesn't depend on altitude like atmospheric drag does, so it affects satellites wherever they orbit around Earth, including high altitudes and geostationary orbits. For communication satellites with large solar panels, this pressure can cause small but continuous changes, mainly affecting the shape of the orbit (eccentricity) and the orientation of the closest point in the orbit (longitude of perigee). In low Earth orbit (LEO), the effect of solar radiation pressure is relatively small compared to atmospheric drag, which dominates because of the denser air at those altitudes. Additionally, the constant movement through Earth's shadow and changes in orientation reduce the overall impact in LEO. But for satellites higher up, where there's almost no atmosphere, solar radiation pressure becomes a significant and persistent factor that must be considered for accurate orbit prediction and satellite control.

2. Luni-solar Gravitational Attractions

The orbital inclination of a satellite gradually changes over time due to the gravitational pull of the Sun and the Moon. The Earth orbits the Sun in an elliptical path, with its orbital plane tilted by about 7° relative to the Sun's equatorial plane. In addition, Earth's rotation axis is tilted by roughly 23° with respect to the perpendicular of this orbital plane, known as the ecliptic. The Moon, meanwhile, revolves around Earth with an inclination of about 5° to Earth's equatorial plane. Because of these combined gravitational influences, a satellite experiences various out-of-plane forces that alter its orbital inclination. For example, these forces can gradually increase a satellite's inclination from 0° up to about 14.67° over time. A noticeable shift in inclination due to the Sun and Moon often called the lunisolar effect takes place over a period of roughly 26.6 years. In practice, however, this isn't a major concern, since most satellites don't operate for that long. Moreover, the Sun's and Moon's gravitational effects are negligible for satellites in low Earth orbit (LEO), where atmospheric drag has a far greater influence on orbital behaviour.

3. Effect Due to Bulged Equator (J22 effect)

Earth's equator isn't perfectly circular—it's slightly elliptical known as J₂₂, with a small eccentricity around 10^{-5} . This ellipticity creates a gravity gradient across Earth's equator, which strongly affects **geostationary satellites**. Ideally, a satellite in geostationary orbit stays fixed above one spot on the equator. However, the gravity differences tied to the ellipticity make satellites drift towards two "stable points" located near the minor axis of Earth's equatorial ellipse—specifically, around 75° E and 105° W longitude. Satellites that are no longer actively controlled eventually drift to these spots, which are often called "satellite graveyards." For operational satellites, station-keeping manoeuvres prevent them from drifting into these graveyards. When a satellite reaches the end of its lifespan, operators deliberately move it into a higher orbit—called a graveyard orbit—about 300 km above the regular geostationary belt. This keeps retired satellites away from active ones and ensures they remain in a stable position without drifting back. The effect of this equatorial ellipticity mostly concerns geostationary satellites; it's negligible for satellites in other types of orbits.

4. Earth's Albedo Effects

Earth's albedo refers to the fraction of incoming solar radiation that is reflected back into space by the planet's surface and atmosphere. This reflected radiation imparts a small but continuous force on a satellite, similar to—but weaker than—direct solar radiation pressure. In orbital dynamics, albedo effects introduce small periodic perturbations, primarily influencing the satellite's eccentricity and argument of perigee. The magnitude of this force depends on factors such as Earth's reflectivity distribution, cloud coverage, and the satellite's position relative to illuminated regions. Although relatively minor compared to J_2 and atmospheric drag, albedo-induced accelerations can become relevant for missions requiring high-precision orbit control.

5. Tidal Effects

Tidal effects come from the constant gravitational pull of the Moon and the Sun on Earth. This pull slightly stretches Earth and shifts its mass distribution, creating tidal bulges that change with time. Because the gravity field is constantly adjusting, satellites feel tiny variations in the force acting on them. These tidal variations introduce slow, long-period changes in the orbit especially in the semi-major axis, eccentricity, and the precession of the orbital plane. The effect is small but important for very accurate orbit modelling or long missions where tiny changes accumulate over time.

CONCLUSION

Result 1: Computational results of Orbital elements in MATLAB

If the initial state vectors of a satellite are available, the corresponding Classical Orbital Elements can be determined using the MATLAB code provided in Appendix A.

Result 2: Computational Result of Ideal Keplerian Orbit in MATLAB

The true anomaly is propagated with time to trace the satellite's trajectory and to compute the state vectors at any desired instant. For an ideal Keplerian orbit, it is observed that both the state vectors and the orbital elements remain constant over time, reflecting the absence of perturbing effects.

Result 3: Computational Result considering J₂ Perturbations

When the J₂ perturbation is introduced into the ideal Keplerian orbit, noticeable variations occur in the Right Ascension of the Ascending Node (RAAN), the argument of perigee, and the mean anomaly. As a result, the state vectors obtained under J₂ perturbation differ from those of the ideal orbit.

For the E08 satellite, the effect of J₂ leads to a change of approximately -12.5° in RAAN and 17° in the argument of perigee when compared to the ideal case. Similarly, for the E06 satellite, the change in RAAN is about -6° , while the argument of perigee varies by around 1.9° .

The influence of the J₂ effect is more pronounced for the E08 satellite due to its lower orbital altitude. In the case of E08, operating at an altitude of 450 km, the difference between the computed results and the STK values is on the order of 1.46 km. In contrast, for E06, which operates at a higher altitude of 740 km, the corresponding difference is smaller, at approximately 1.34 km. Also, the difference in velocity magnitude of E08 is of order 0.00164 km/s, whereas for E06 it is 0.00142 km/s.

Result 4: Computational Result considering both J₂ and Drag

When atmospheric drag is included, it leads to a reduction in the semi-major axis, which in turn alters the satellite's mean motion. This change indirectly influences the evolution of the RAAN and the argument of perigee. The STK software does not provide an option to model only J₂ and drag; instead, it incorporates higher-order gravitational effects such as J₃, J₄, J₂₂, and others. Consequently, the larger discrepancies observed between the MATLAB-computed results and the STK outputs can be attributed to the simplified perturbation model adopted in this work.

For the E08 satellite, the combined effect of J₂ and atmospheric drag results in a change of approximately -6.2° in RAAN and 8.5° in the argument of perigee when compared to the ideal orbit. In the case of E06, the change in RAAN is about -6° , while the argument of perigee varies by around 1.9° .

For E08, the semi major axis decay occurring due to drag for each day is about -0.07km.

The resulting position differences are observed to be on the order of 130.9 km for E08 and approximately 1068 km for E06. The velocity magnitude difference for E08 is of order 0.1431 km/s, and for E06 it is of order 1.12 km/s.

As part of the future scope, this work can be extended to include higher-order Earth gravity models in order to improve the accuracy and achieve closer agreement with high-fidelity tools such as STK.

The Computation results validate that:

- **J₂** alone introduces predictable, periodic variations with small discrepancies.
- **J₂ + drag** produces cumulative changes, most notably semi-major axis decay, leading to increasing differences in both position and orbital elements.
- **Lower orbits (E08)** experience stronger J2 perturbation deviations than higher orbit ones (E06).
- **Lower orbits (E08)** stronger decay in semi major axis thereby reducing the orbital period of the satellite than higher orbit ones (E06)

These comparisons demonstrate that accurately modelling long-term satellite motion requires including both geopotential and drag effects, especially for LEO satellites where orbital decay becomes significant.

FUTURE SCOPE OF WORK

This work is limited to the two dominant perturbations affecting Low Earth Orbit:

- **J_2 gravitational perturbation**, representing the effect of Earth's oblateness on the secular evolution of orbital elements.
- **Atmospheric drag**, the primary non-conservative force in LEO, responsible for long-term orbital energy loss and altitude decay.

The future scope of this project work will be focused on including all the Gravity models.

For future expansion, additional perturbing effects may be incorporated, such as:

- **Higher-order Earth gravity harmonics ($J_3, J_4, \dots$)**
- **Solar radiation pressure**
- **Third-body gravitational influences from the Moon and the Sun**
- **Earth albedo effects**
- **Tidal forces**

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